The subsequent decline occurs more slowly over several months, and its rate of decline defines the **speed class** of a nova. A **fast nova** takes a few weeks to dim by two magnitudes, whereas a **slow nova** may take nearly 100 days to decline by the same amount from maximum; see Figs. 18.17 and 18.18. The declines are sometimes punctuated by large fluctuations in brightness, which in extreme cases may take the form of the complete absence of visible light from the nova for a month or so before it reappears. Fast novae are typically three magnitudes brighter than slow novae, but in either case a nova falls to nearly its pre-eruption appearance after a few decades.

During the first few months, the decline in brightness occurs only at visual wavelengths. When observations at infrared and ultraviolet wavelengths are included, the bolometric

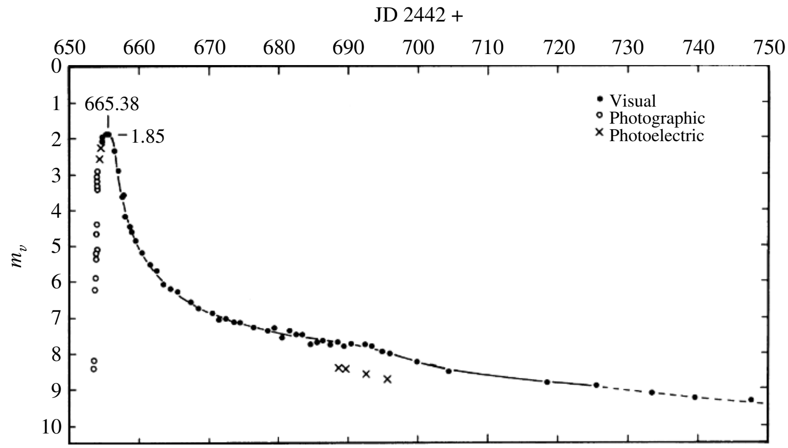

FIGURE 18.17 The light curve of V1500 Cyg, a fast nova. (Figure adapted from Young, Corwin, Bryan, and De Vaucouleurs, *Ap. J.*, *209*, 882, 1976.)

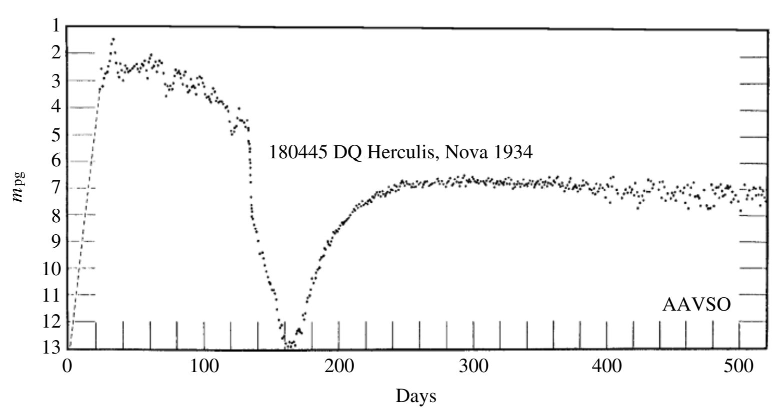

FIGURE 18.18 The light curve of DQ Her, a slow nova. The *photographic magnitude*, m_{pg}, is measured from the nova's image on photographic plates. (We acknowledge with thanks the variable star observations from the AAVSO International Database contributed by observers worldwide.)

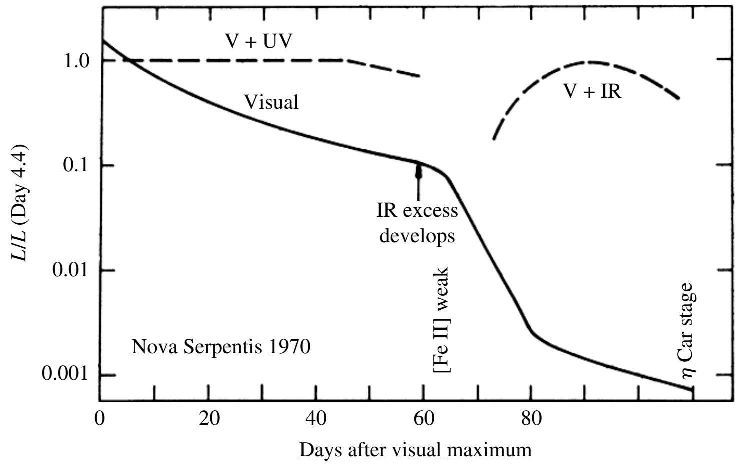

FIGURE 18.19 The bolometric luminosity of nova FH Serpentis, in terms of its luminosity at day 4.4. Note that during the first 60 days, the decline in visible energy was almost exactly offset by an increase at ultraviolet wavelengths. Thereafter, the infrared rose as the visible light output was redistributed to infrared wavelengths. (Figure adapted from Gallagher and Starrfield, *Annu. Rev. Astron. Astrophys.*, *16*, 171, 1978. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 16, ©1978 by Annual Reviews Inc.)

luminosity of a nova is found to remain approximately constant for several months following its outburst; see Fig. 18.19. In addition, spectra of novae show that they are accompanied by the ejection of 10^{-5} to 10^{-4} $M_\odot$ of hot gases at velocities between several hundred and several thousand km s^{-1}. The speed of the gases is roughly three times greater for fast novae, but the total mass ejected is about the same for both speed classes. We will see that the changing characteristics of this expanding shell of gas are responsible for the features seen in Fig. 18.19.

The average value of the absolute visual magnitude of a nova in its quiescent state is $M_V = 4.5$. Assuming that the light from such a system comes primarily from the accretion disk around the white dwarf, an estimate of the mass transfer rate for a typical nova can be obtained. (For the purposes of this estimation, visual magnitudes will be used instead of bolometric magnitudes. This means that the mass transfer rate will be slightly underestimated.) From Eq. (3.7), the luminosity of the system is

$$L = 100^{(M_{\rm Sun} - M_V)/5}\ L_\odot = 1.3\ L_\odot = 4.9 \times 10^{26}\ \text{W}.$$

With this result, the luminosity of an accretion disk (Eq. 18.23) can be solved for the mass transfer rate, giving

$$\dot{M} = \frac{2RL}{GM} = 5.7 \times 10^{13}\ \text{kg s}^{-1},$$

or about $9.0 \times 10^{-10}\ M_\odot\ \text{yr}^{-1}$.

This is in good agreement with the accepted theoretical model of a nova, which incorporates a white dwarf in a semidetached binary system that accretes matter at a rate of about 10^{-8} to 10^{-9} $M_\odot$ yr^{-1}. The hydrogen-rich gases accumulate on the surface of the white dwarf, where they are compressed and heated. At the base of this layer, turbulent mixing enriches the gases with the carbon, nitrogen, and oxygen of the white dwarf. (Without this mixing, the ensuing explosion would be too feeble to eject the mass observed for the expanding shell of hot gases.) Spectroscopic analysis of the shell shows an enrichment of carbon, nitrogen, and oxygen by a factor of 10 to 100 times the solar abundance of these elements.

At the base of this enriched layer of hydrogen, the material is supported by electron degeneracy pressure. When about 10^{-4} to 10^{-5} $M_\odot$ of hydrogen has accumulated and the temperature at the base reaches a few million kelvins, a shell of CNO-cycle hydrogen burning develops. For highly degenerate matter the pressure is independent of the temperature, so the shell source cannot dampen the reaction rate by expanding and cooling. The result is a runaway thermonuclear reaction, with temperatures reaching 10^8 K before the electrons lose their degeneracy.[20] When the luminosity exceeds the Eddington limit of about 10^{31} W (recall Eq. 10.114), radiation pressure can lift the accreted material and expel it into space. The fast and slow speed classes of novae are likely due to variations in the mass of the white dwarf and in the degree of CNO enrichment of the hydrogen surface layer. The brief standstill that occurs before maximum luminosity is probably an effect of the changing opacity of the ejecta.

The energy that would be released in the complete fusion of a hydrogen layer of $m = 10^{-4}\ M_\odot$ is $0.007mc^2 \approx 10^{41}$ J, roughly 10^3 times larger than the energies actually observed. If all of the hydrogen were in fact consumed, the nova would shine for several hundred years. Most of the accumulated material must therefore be propelled into space by the explosion. However, the kinetic energy of the ejecta (far from the nova) is much smaller than the gravitational binding energy of the surface layer, indicating that the total energy given to the ejecta is just barely enough to allow it to escape from the system.

Only about 10% of the hydrogen layer is ejected by the nova explosion. Following this initial **hydrodynamic ejection phase** which dominates for fast novae, hydrostatic equilibrium is established and the **hydrostatic burning phase** begins. During this prolonged stage of hydrogen burning, which is most important for slow novae, energy is produced at a constant rate approximately equal to the Eddington luminosity. The layer above the shell of CNO burning becomes fully convective and expands by a factor of 10 to 100, extending to some 10^9 m.[21] At the surface of the convective envelope the effective temperature is about 10^5 K, much less than the 4×10^7 K in the active CNO shell source below.

Finally, the last of the accreted surface layer is ejected, from between a few months to about a year after the hydrostatic burning phase began. Deprived of fuel, the hydrostatic burning phase ends, and the white dwarf begins to cool. Eventually the binary system reverts to its quiescent configuration and the accretion process begins anew. For accretion rates of

[20]This mechanism is similar to the helium core flash that was described in Section 13.2.

[21]The white dwarf remnant may overflow its Roche lobe. The consequences of the resulting disruption of the close binary system are not yet clear.

10^{-8} to 10^{-9} $M_\odot$ yr^{-1}, it will take some 10^4 to 10^5 years to build up another surface layer of 10^{-4} $M_\odot$.

The physical character of the *ejected gases* passes through three distinct phases as a consequence of the nova explosion. During the initial **fireball expansion phase**, the material blown off the star in the hydrodynamic ejection phase forms an optically thick "fireball" that radiates as a hot blackbody of 6000–10,000 K. The observed light originates in the "photosphere" of the expanding fireball; at this point, the spectrum of the nova resembles that of an A or F supergiant.

The fireball expansion phase will be examined in Problem 18.13, where you will consider a simple model of a nova for which mass is ejected at a constant rate of $\dot{M}_{\rm eject}$ at a constant speed v. The expanding model photosphere has a radius that initially increases linearly with time and then approaches a limiting value of

$$R_\infty = \frac{3\overline{\kappa}\dot{M}_{\rm eject}}{8\pi v}. \tag{18.30}$$

If the luminosity, L, of the nova is also assumed to be constant, then from Eq. (3.17) the effective temperature of the model photosphere approaches

$$T_\infty = \left(\frac{L}{4\pi\sigma}\right)^{1/4}\left(\frac{8\pi v}{3\overline{\kappa}\dot{M}_{\rm eject}}\right)^{1/2}. \tag{18.31}$$

For an opacity of $\overline{\kappa} = 0.04$ m^2 kg^{-1} [Eq. (9.27) for electron scattering, with pure hydrogen assumed for convenience], a mass ejection rate of $\dot{M}_{\rm eject} \approx 10^{19}$ kg s^{-1} (about 10^{-4} $M_\odot$ yr^{-1}), and an ejection speed of $v \approx 1000$ km s^{-1}, the fireball's photosphere approaches a limiting radius of about 5×10^{10} m, or 1/3 AU. Taking L to be the Eddington limit of about 10^{31} W, the effective temperature of the model photosphere approaches a value of nearly 9000 K.

The optically thick fireball phase ends in a few days, at the point of maximum visual brightness. Then, as the shell of gas thrown off by the nova continues to expand, it becomes less and less dense. The rate of mass ejection, $\dot{M}_{\rm eject}$, has also declined in the hydrostatic burning phase. The result, according to Eqs. (18.30) and (18.31), is that the location of the photosphere moves inward and its temperature increases slightly. Although these general trends are correct, the opacity is in fact very sensitive to the temperature for $T < 10^4$ K (recall Fig. 9.10), and our model is too simplistic to describe the evolution of the nova. More advanced arguments show that as the visual brightness declines, more light is received from the nova at ultraviolet wavelengths. Finally, the shell becomes transparent and the **optically thin phase** begins. The central white dwarf, swollen by its hydrostatic burning phase, now has the appearance of a blue horizontal-branch object located just blueward of the RR Lyrae stars on the H–R diagram. The white dwarf envelope may burn irregularly, resulting in the substantial fluctuations in brightness observed for some novae.

After a few months, when the temperature of the expanding envelope of gases has fallen to about 1000 K, carbon in the ejecta can condense to form dust consisting of graphite grains.[22]

[22]The identification of the grain composition comes in part from an infrared emission "bump" at a wavelength of 5 μm; recall Section 12.1. Novae are natural laboratories for testing theories of grain formation.

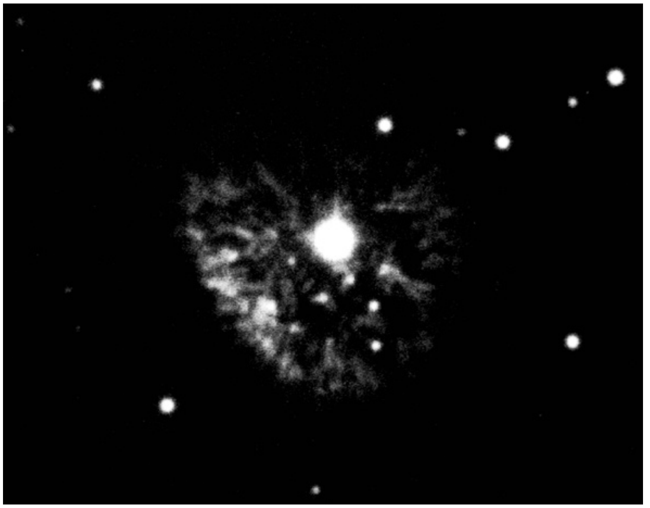

FIGURE 18.20 A 1949 photo of Nova Persei, which exploded in 1901. (Courtesy of Palomar/Caltech.)

This initiates the **dust formation phase**. The resulting dust shell becomes optically thick in roughly 50% of all novae. The visible light from a nova is undiminished by an optically thin shell, but the formation of an optically thick cocoon of dust obscures or completely hides the central white dwarf. In the latter case, the output of visible light suddenly plunges, as seen in Fig. 18.19. The light from the white dwarf is absorbed and re-emitted by the graphite grains, so the optically thick dust shell radiates as a ~ 900 K blackbody at infrared wavelengths. In this way, the nova's bolometric luminosity remains constant as long as the white dwarf continues to produce energy at roughly the Eddington rate while in its hydrostatic burning phase. Figure 18.20 shows that the expanding shell may remain visible for years after the hydrostatic burning phase has ended, its gases and dust enriching the interstellar medium.

Polars: X-Rays from White Dwarf Systems

AM Herculis stars (also called **polars**), are semidetached binaries containing white dwarfs with magnetic fields of about 2000 T. The torque produced by the white dwarf's field interacting with the secondary star's envelope results in a nearly synchronous rotation; the two stars perpetually face each other, connected by a stream of hot gas.[23] As this gas approaches the white dwarf, it moves almost straight down toward the surface and forms an accretion *column* a few tens of kilometers across. A shock front occurs above the white dwarf's photosphere, where the gas is decelerated and heated to a temperature of several 10^8 K. The hot gas emits hard X-ray photons; some escape, and some are absorbed by the photosphere and re-emitted at soft X-ray and ultraviolet wavelengths.

[23] If the white dwarf has a somewhat weaker field ($B_s < 1000$ T), or if the stars are farther apart, an accretion disk may form, only to be disrupted near the star (as shown in Fig. 18.22). These systems, called **DQ Herculis stars**, or **intermediate polars**, do not exhibit synchronous rotation.

The visible light observed from these systems is in the form of *cyclotron radiation* emitted by nonrelativistic electrons spiraling along the magnetic field lines of the accretion column. This is the nonrelativistic analog of the synchrotron radiation emitted by relativistic electrons; see Fig. 16.17. In contrast to the continuous spectrum of synchrotron radiation, most of the energy of cyclotron radiation is emitted at the *cyclotron frequency*,

$$\boxed{\nu_c = \frac{eB}{2\pi m_e}.} \tag{18.32}$$

For $B_s = 1000$ T, $\nu_c = 2.8 \times 10^{13}$ Hz, which is in the infrared. However, a small fraction of the energy is emitted at higher harmonics (multiples) of ν_c and may be detected at visible wavelengths by astronomers on Earth. The cyclotron radiation is circularly polarized when observed parallel to the direction of the magnetic field lines, and linearly polarized when viewed perpendicular to the field lines.[24] Thus, as the two stars orbit each other (typically every 1 to 2 hours), the measured polarization changes smoothly between being circularly and linearly polarized. In fact, it is this strong variable polarization (up to 30%) that gives polars their name.

18.5 ■ TYPE IA SUPERNOVAE

We have seen that there are many differences among the characteristics of individual novae. The peak luminosity, the rate of decline, the presence of rapid fluctuations, and/or the complete disappearance of the nova at visible wavelengths—all of these vary greatly from system to system. On the other hand, another type of cataclysmic variable, the **Type Ia supernova**, varies relatively little and in a systematic way. This means that it is possible to use these exploding stars as calibrated luminosity sources (“standardizable candles”), allowing astronomers to establish the distances to the systems in which they are found.[25]

Observations

Type Ia supernovae are remarkably consistent in their energy output; at maximum light most Type Ia’s reach an average maximum in the blue and visual wavelength bands of

$$\langle M_B \rangle \simeq \langle M_V \rangle \simeq -19.3 \pm 0.3,$$

with a typical spread of less than about 0.3 magnitudes. As can be seen in Fig. 18.21, a clear relationship exists between the peak brightness and the rate of decline in the light curve (the brightest Type Ia’s decline the slowest), making it possible to accurately determine the maximum luminosity of an individual Type Ia by measuring the rate of decline. Knowing the luminosity (or absolute magnitude), we can compute the distance to the supernova. Given their tremendous brightness, Type Ia supernovae serve as critically important tools

[24]The electric field vector of linearly polarized light oscillates in a single plane, whereas for circularly polarized light this plane of polarization rotates about the direction of travel.
[25]Core-collapse supernovae (Types Ib, Ic, and II) were discussed in detail in Chapter 15.

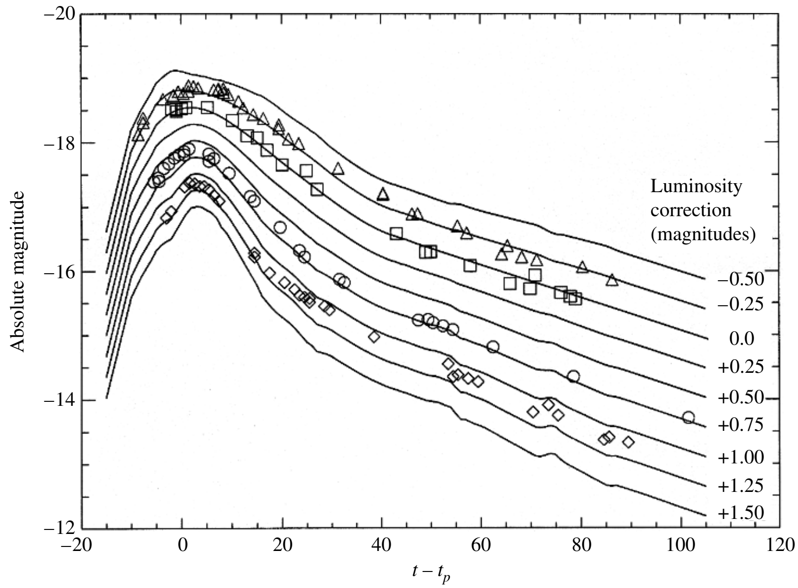

FIGURE 18.21 The rate of decline in the light curve of a Type Ia supernova is inversely correlated with the maximum brightness of the light curve. (Figure adapted from Riess, Press, and Kirshner, *Ap. J.*, *438*, L17, 1995.)

for measuring the distances to the galaxies in which they reside. This in turn means that astronomers can probe the structure of the universe to great distances. In fact, Type Ia supernovae played a crucial role in demonstrating that the expansion of the universe is actually accelerating today, 13.7 billion years after the Big Bang, and that nearly three-fourths of the universe consists of **dark energy**.

As was mentioned in Section 15.2, Type Ia supernovae do not exhibit hydrogen lines in their spectra and, instead, show the strong presence of Si II lines, along with neutral and ionized lines of O, Mg, S, Ca and Fe (Fig. 15.6). Given that hydrogen is the most abundant element in the universe, the absence of hydrogen indicates that Type Ia supernovae are evolved objects that have either lost their hydrogen or had it converted to heavier elements, or both. The spectral lines also show P-Cygni profiles, representative of mass loss. In addition, the blueshifted absorption features indicate expansion velocities of the ejecta of $\gtrsim 10^4$ km s^{-1} ($\gtrsim 0.03c$).

Models of Type Ia Supernovae

Given the remarkable consistency of Type Ia light curves and spectra, it appears that a fairly uniform mechanism must be responsible for these extremely energetic events.

The standard model for Type Ia supernovae assumed by astronomers today is that these events are due to the destruction of a white dwarf star in a binary system. If sufficient mass falls onto the white dwarf, its mass can be driven to near the Chandrasekhar limit, producing a catastrophic explosion. Still unclear at the time this text was written is the exact mechanism (or mechanisms) that trigger the explosion.

Two general scenarios have been proposed. In one scenario, known as **double-degenerate** models, two white dwarf stars exist in a binary orbit. One of the most dramatic predictions of Einstein's general theory of relativity is the existence of **gravitational waves** (or **gravitational radiation**). According to general relativity, mass acts on spacetime, telling it how to curve. If the distribution of a system's mass varies, the resulting changes in the surrounding spacetime curvature may propagate outward as a gravitational wave, carrying energy and angular momentum away from the system. (If the collapse of a star is spherically symmetric, it will not produce gravitational waves; there must be a departure from spherical symmetry.) When applied to a close binary system, general relativity shows that the emission of gravitational radiation will cause the stars to spiral together. If the orbital period is under about 14 hours, the loss of energy via gravitational waves governs the subsequent evolution of a system with solar-mass components. For example, as a white dwarf and a neutron star spiral closer together, the white dwarf may break up and donate some of its mass and angular momentum to its companion. The result could be an isolated millisecond pulsar. A system of two neutron stars, known as the Hulse–Taylor pulsar, has confirmed this prediction of general relativity to incredibly high precision; see the extended discussion beginning on page 703.

In the case where two white dwarf stars are spiraling together, the less massive star (which has the larger radius) will eventually spill over its Roche lobe and be completely torn apart in just a few orbits. The resulting thick disk dumps its C–O-rich material onto the more massive primary. As the mass of the primary grows and nears the Chandrasekhar limit, nuclear reactions begin in the deep interior, eventually destroying the primary white dwarf (this scenario was illustrated in Fig. 18.11).

Double-degenerate models appear to predict about the right number of mergers, consistent with the observed Type Ia supernova rate in galaxies, and they naturally account for the lack of hydrogen in the spectra of Type Ia's. However, computer simulations of nuclear burning suggest that the ignition may be off-center, resulting in ultimate collapse to a neutron star, rather than complete disruption of the white dwarf as a supernova. In addition, it appears that the production of heavy elements may be inconsistent with the relative abundances observed in supernova spectra.

The other general scenario, known as **single-degenerate** models, involves an evolving star in orbit about a white dwarf, much like the models of dwarf novae and novae. However, in this case, the mass falling onto the white dwarf results in complete destruction of the white dwarf in a Type Ia supernova. To date, this set of models is generally favored, but the details of the eruption are still unclear.

One version of single-degenerate models suggests that as the material from the secondary falls onto the primary, the helium in the gas will settle on top of the C–O white dwarf, becoming degenerate. When enough helium has accumulated, a helium flash will occur. Not only will this cause the helium to burn to carbon and oxygen, but it will also send a shock wave downward into the degenerate C–O white dwarf, causing ignition of the degenerate carbon and oxygen.

A second version of the single-degenerate models doesn't invoke degenerate helium burning on the surface but simply has carbon and oxygen igniting in the interior of the white dwarf as the star nears the Chandrasekhar limit, at which point the degenerate gas is no longer able to support the mass of the star. As the star approaches the fatal limit, two-and three-dimensional simulations suggest that multiple, independent ignition points may occur deep within the core, resulting in nonspherical events.

What happens next is also a matter of significant debate and ongoing research. It is as yet unclear if the resulting burning front of carbon and oxygen occurs at subsonic speeds (known as a **deflagration** event) or if the front accelerates and steepens to become a supersonic burning front (known as a **detonation**, or a true explosion). Precisely how the burning front advances affects the details of the resulting light curve (the maximum luminosity and rate of decay following maximum) as well as the relative abundances of the elements produced as observed in the spectrum. Of course, the question of deflagration versus detonation applies to successful double-degenerate models as well.

In all versions of the single-degenerate scenario, one of the general challenges has been to have just the right rate of accretion from the secondary. If the accretion rate isn't appropriately fine-tuned, the result could be a dwarf nova or a classical nova.

It is possible that both double- and single-degenerate mechanisms may be at work in nature. It is also possible that some single-degenerate events (if they occur) may invoke helium flashes while others may simply ignite carbon and oxygen in the interior without the helium trigger. Perhaps even deflagration and detonation events occur. In any case, the consistency of the light curves ultimately arises from the eruption of a C–O white dwarf near 1.4 $M_\odot$. The variations may arise from slight variations in mass and/or variations in mechanisms.

Much work remains to be done in understanding Type Ia supernovae, which are so critically important to so many aspects of modern astrophysics.

18.6 ■ NEUTRON STARS AND BLACK HOLES IN BINARIES

If one of the stars in a close binary system is sufficiently massive that it explodes as a core-collapse supernova, the result may be either a neutron star or a black hole orbiting the companion star. In a semidetached system, hot gas can then spill through the inner Lagrangian point from the distended atmosphere of the companion star onto the compact object. A variety of intriguing phenomena are powered by the energy released when the gas falls down the deep gravitational potential well onto the compact object. As will be seen shortly, many of these systems emit copious quantities of X-rays. In fact, these **binary X-ray systems** shine most strongly in the X-ray region of the electromagnetic spectrum. Other systems may consist of *two* compact objects, such as the binary pulsars.

Formation of Binaries with Neutron Stars or Black Holes

Whether or not a binary system survives the supernova explosion of one of its component stars depends on the amount of mass ejected from the system.[26] Consider a system initially containing two stars of mass M_1 and M_2, separated by a distance a, that are in circular orbits about their common center of mass. Using Eq. (2.35), we find that the total energy of the

[26] We have also seen that asymmetric jets during the formation of a neutron star may give the neutron star a violent kick, which would disrupt the system; recall Section 16.7.

system is

$$E_i = \frac{1}{2}M_1 v_1^2 + \frac{1}{2}M_2 v_2^2 - G\frac{M_1 M_2}{a} = -G\frac{M_1 M_2}{2a}. \tag{18.33}$$

The speeds of the two stars are related by Eq. (7.4), $M_1 v_1 = M_2 v_2$. Now suppose that Star 1 explodes as a core-collapse supernova, leaving a remnant of mass M_R. For a spherically symmetric explosion, there is no change in the velocity of Star 1. Before the spherical shell of ejecta reaches Star 2, its mass acts gravitationally as though it were still on Star 1 (recall Example 2.2.1). So far, the supernova has had no effect on the binary. However, as soon as the shell has swept beyond Star 2, the gravitational influence of the ejecta is no longer detectable.

Thus the main consequence of the supernova on the orbital dynamics of the binary system arises from the ejection of mass, the removal of some of the gravitational glue that was binding the stars together.[27] Since the velocity of Star 2 is initially unchanged and the separation of the two stars remains the same, the total energy of the system after the explosion is now

$$E_f = \frac{1}{2}M_R v_1^2 + \frac{1}{2}M_2 v_2^2 - G\frac{M_R M_2}{a}. \tag{18.34}$$

If the explosion results in an *unbound* system, then $E_f \geq 0$. It is left as an exercise to show that the mass of the remnant must satisfy

$$\frac{M_R}{M_1 + M_2} \leq \frac{1}{(2 + M_2/M_1)(1 + M_2/M_1)} < \frac{1}{2} \tag{18.35}$$

for an unbound system. That is, *at least* one-half of the total mass of the binary system must be ejected if the supernova explosion of Star 1 is to disrupt the system. If one-half or more of the system's mass is retained, the result will be a neutron star or a black hole gravitationally bound to a companion star. For a massive companion star ($M_2 \gg M_1$), this is a likely result.

Capturing Isolated Neutron Stars

It is possible that isolated neutron stars, formed by core-collapse supernovae, may be gravitationally captured during a chance encounter with another star. Because the total energy of two unbound stars is initially greater than zero, some of the excess kinetic energy must be removed for a capture to occur.

If the proximity of the two objects raises a tidal bulge on the nondegenerate star, energy may be dissipated by the damping mechanisms discussed in Section 14.2 for pulsating stars. The outcome of such a **tidal capture** depends on the nearness of the passage and the type of star involved. If the neutron star passes between about 1 to 3 times the radius of the other star, the resulting binary system will have a period ranging from several hours (with a main-sequence star) to several days (with a giant).

[27]The direct impact of the supernova blast on the companion star has been neglected, although this too will contribute to disrupting the system.

This tidal capture process is most effective in regions that are extremely densely populated with stars, such as the centers of globular clusters (Section 13.3). It is estimated that in a compact globular cluster, tidal capture could produce up to about ten close binary systems containing a neutron star over a period of some 10^{10} years. This is consistent with the number of X-ray sources observed in globular clusters. (The estimated lifetime of a binary X-ray system is on the order of 10^9 years, so only the most compact globular clusters would be expected to harbor even one X-ray source at a given time; see Problem 18.20.)

An alternative capture mechanism involves three (or more) stars. One of the stars would be gravitationally flung from the system, removing energy and so allowing the capture to take place.

Yet another possibility was envisioned by by Kip Thorne and Anna Żytkow of Caltech in 1977. Although a direct hit would destroy a main-sequence star, the penetration of a neutron star into a giant star would bring it close to the star's degenerate core. The result could be a neutron star orbiting inside the giant star: a system that is known as a **Thorne–Żytkow object**. It is thought that the envelope of the giant star would be quickly expelled, producing a neutron star–white dwarf binary with an orbital period of about 10 minutes. (To date, these objects remain hypothetical.)

Binary X-Ray Pulsars

Close binary systems containing neutron stars were first identified by their energetic emission of X-rays. The first source of X-rays beyond the Solar System was discovered in 1962 in the constellation Scorpius by a Geiger counter arcing above Earth's atmosphere in a sounding rocket. (X-rays cannot penetrate the atmosphere, so detectors and telescopes designed for X-ray wavelengths must make their observations from space.[28]) This object, called Sco X-1, is now known to be a **binary X-ray pulsar** (also called simply an X-ray pulsar). The periodic eclipse of another X-ray pulsar, Cen X-3 in the constellation Centaurus, revealed its binary nature. [It is important to note that the Crab pulsar also emits X-rays (along with a small number of other isolated pulsars), but the Crab is primarily a radio pulsar that radiates in every region of the electromagnetic spectrum.]

X-ray pulsars are powered by the gravitational potential energy released by accreting matter. Recall from Example 18.1.2 that when mass falls from a great distance to the surface of a neutron star, about 20% of its rest energy is released, an amount that far exceeds the fraction of a percent that would be produced by fusion. The observed X-ray luminosities range up to 10^{31} W [the Eddington limit; see Eq. (10.6)]. For a neutron star with a radius of 10 km, the Stefan–Boltzmann equation in the form of Eq. (3.17) shows that the temperature associated with this luminosity is about 2×10^7 K. According to Wien's law, Eq. (3.19), the spectrum of a blackbody with this temperature would peak at an X-ray wavelength of about 0.15 nm.

X-ray pulsars also emit radio wavelength energy, just like isolated pulsars. However, radio wavelength emissions are easily squelched by the accretion disk in the binary system, and so the radio emissions are not as prominant as they are for isolated pulsars.

[28]The first X-ray detector was designed to look for X-rays from the lunar surface, produced when solar wind particles cause the lunar soil to fluoresce. The presence of enormously stronger cosmic X-ray sources came as a surprise to astronomers at the time.

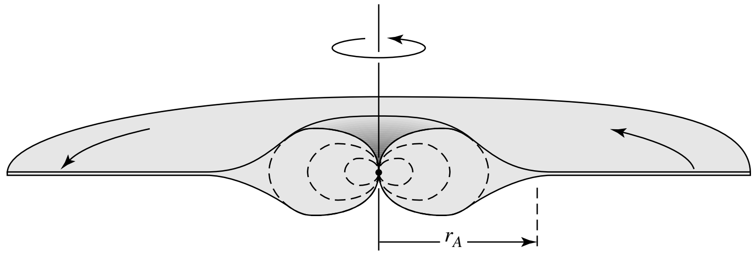

FIGURE 18.22 Accreting gas channeled onto a neutron star's magnetic poles, where $r \approx r_A$.

You should recall from Section 16.6 that neutron stars are often accompanied by powerful magnetic fields. In fact, these fields may be sufficiently strong to prevent the accreting matter from even reaching the star's surface. The strength of the neutron star's magnetic dipole field is proportional to $1/r^3$, so the plunging gases encounter a rapidly increasing field. When the magnetic energy density $u_m = B^2/2\mu_0$ (Eq. 11.9) becomes comparable to the kinetic energy density $u_K = \frac{1}{2}\rho v^2$, the magnetic field will channel the infalling ionized gases toward the poles of the neutron star; see Fig. 18.22. This occurs at a distance from the star known as the **Alfvén radius**, r_A, where

$$\frac{1}{2}\rho v^2 = \frac{B^2}{2\mu_0}. \tag{18.36}$$

For the special case of spherically symmetric accretion, with the gases starting at rest at a great distance, energy conservation implies that the free-fall velocity is $v = \sqrt{2GM/r}$ for a star of mass M. Furthermore, the density and velocity are related to the mass accretion rate, $\dot{M}$, by Eq. (11.3),

$$\dot{M} = 4\pi r^2 \rho v, \tag{18.37}$$

and the radial dependence of the magnetic dipole field strength may be expressed as

$$B(r) = B_s \left(\frac{R}{r}\right)^3, \tag{18.38}$$

where B_s is the surface value of the magnetic field. Inserting these expressions into Eq. (18.36) and solving for the Alfvén radius, we obtain

$$r_A = \left(\frac{8\pi^2 B_s^4 R^{12}}{\mu_0^2 G M \dot{M}^2}\right)^{1/7} \tag{18.39}$$

(the proof is left as an exercise). Of course, the accretion will not actually be spherically symmetric. However, the magnetic field increases so rapidly as the falling matter approaches the star that a more realistic calculation yields nearly the same result: The flow will be disrupted at a *disruption radius* r_d,

$$r_d = \alpha r_A, \tag{18.40}$$

with $\alpha \sim 0.5$.

Example 18.6.1. Before considering the details of channeled accretion onto a neutron star, let's look at the case of accretion onto the white dwarf considered in Example 18.2.1, for which $M = 0.85\ \mathrm{M}_\odot$, $R = 0.0095\ \mathrm{R}_\odot = 6.6 \times 10^6$ m, and $\dot{M} = 10^{13}$ kg s^{-1} ($1.6 \times 10^{-10}\ \mathrm{M}_\odot$ yr^{-1}). Assume that its magnetic field has a surface strength of $B_s = 1000$ T, about 100 times stronger than the typical value for a white dwarf. Then, from Eq. (18.39), the Alfvén radius is

$$r_A = 6.07 \times 10^8 \text{ m.}$$

This is comparable to the separation of the stars in a cataclysmic variable (see Example 18.4.1), so an accretion disk cannot form around a white dwarf with an extremely strong magnetic field. Instead, the mass spilling through the inner Lagrangian point is confined to a stream that narrows as it is magnetically directed toward one (or both) of the poles of the white dwarf. In the absence of an accretion disk, all of the accretion energy will be delivered to the pole(s) of the star, with an accretion luminosity of (Eq. 18.24)

$$L_{\mathrm{acc}} = G\frac{M\dot{M}}{R} = 1.71 \times 10^{26} \text{ W.}$$

A Polar Analog in a Neutron Star System

Example 18.6.2. Consider the case of accretion onto the neutron star described in Example 18.2.1. For this star, $M = 1.4\ \mathrm{M}_\odot$, $R = 10$ km, and $\dot{M} = 10^{14}$ kg s^{-1} ($1.6 \times 10^{-9}\ \mathrm{M}_\odot$ yr^{-1}). Furthermore, take the value of the magnetic field at the neutron star's surface to be $B_s = 10^8$ T. The value of the Alfvén radius is then given by Eq. (18.39),

$$r_A = 3.09 \times 10^6 \text{ m.}$$

Although 300 times the radius of the neutron star itself, this is much less than the value of r_{circ} (Eq. 18.25) that describes the extent of an accretion disk. Thus an accretion disk will form around the neutron star but will be disrupted near the neutron star's surface as shown in Fig. 18.22 (unless the magnetic field is quite weak, roughly $< 10^4$ T). As the accreting gas is funneled onto one of the magnetic poles of the neutron star, it forms an accretion column similar to the one described for polars. In this case, however, the accretion luminosity (Eq. 18.24) is four orders of magnitude greater,

$$L_{\mathrm{acc}} = G\frac{M\dot{M}}{R} = 1.86 \times 10^{30} \text{ W,}$$

close to the Eddington limit of $\sim 10^{31}$ W; see Eq. (10.6). As L_{acc} approaches L_{Ed}, radiation pressure elevates the shock front to heights reaching $r \sim 2R$. As a result, X-rays are emitted over a large solid angle.

Eclipsing, Binary X-Ray Pulsar Systems

If the neutron star's magnetic and rotation axes are not aligned, as in Fig. 16.25, the X-ray-emitting region may be eclipsed periodically, and the result is an **eclipsing, binary X-ray pulsar**. Figure 18.23 shows the signal received from Hercules X-1, which exhibits a pulse of X-rays every 1.245 s (the rotation period of the neutron star). Note that the broad pulse (due to the large solid angle of the emission) may occupy $\sim 50\%$ of the pulse period, compared to the sharper radio pulses shown in Fig. 16.19, which take up only 1% to 5% of the pulse period. To date, about 20 binary X-ray pulsars have been found, with periods ranging from 0.15 s to 853 s. As noted on page 590, white dwarfs cannot rotate as rapidly as the lower end of this period range without breaking up. This is one indication that X-ray pulsars are indeed accreting neutron stars.

Further confirmation that most X-ray pulsars are accreting neutron stars comes from the observation that the periods of these objects are slowly decreasing. As time passes, they spin *faster*.[29] The time derivative of the star's rotation period, $\dot{P} \equiv dP/dt$, is related to the rate of change of its angular momentum, $L = I\omega$, by

$$\frac{dL}{dt} = I\frac{d\omega}{dt} = I\frac{d}{dt}\left(\frac{2\pi}{P}\right) = -2\pi I\frac{\dot{P}}{P^2},$$

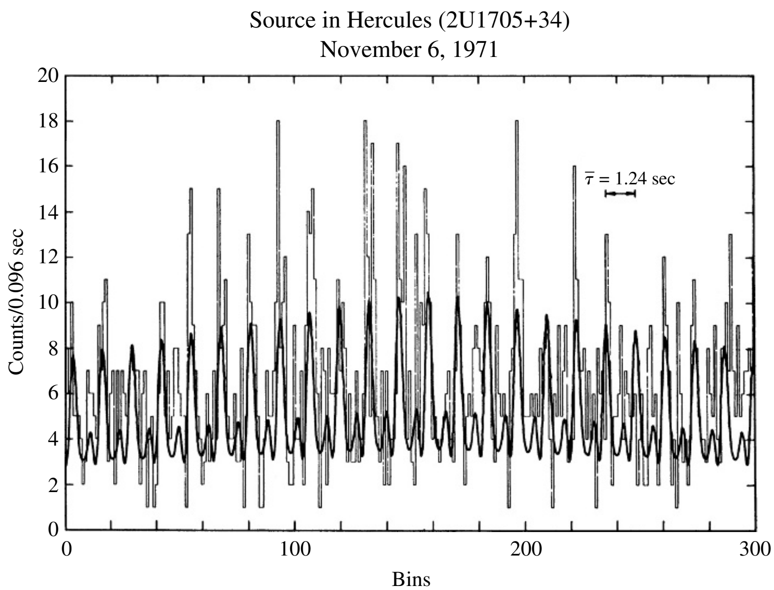

FIGURE 18.23 X-ray pulses from Her X-1, with a period of 1.245 s. The peaks are the X-ray counts received from Her X-1 grouped in bins that are 0.096 s wide, and the heavier curve is a fit to the data using sine functions. (Figure adapted from Tananbaum et al., *Ap. J. Lett.*, *174*, L143, 1972.)

[29] Recall from Section 16.7 that the periods of radio pulsars *increase* with time as they lose energy due to magnetic dipole radiation.

where I is the moment of inertia of the neutron star. Near the disruption radius, the angular momentum of the gas parcels orbiting in the accretion disk ($L = mvr$) is transferred to the neutron star via magnetic torques. The time derivative of the neutron star's angular momentum is just the rate at which angular momentum arrives at the disruption radius, so at $r = r_d$ we set

$$\frac{dL}{dt} = \dot{M} v r_d,$$

where the orbital velocity at $r = r_d$ is $v = \sqrt{GM/r_d}$ [Eq. (2.33) or (2.34) with $e = 0$ and $a = r_d$ for a circular orbit]. Equating these expressions for dL/dt and using the definitions of the Alfvèn and disruption radii, Eqs. (18.39) and (18.40), results in

$$\frac{\dot{P}}{P} = -\frac{P\sqrt{\alpha}}{2\pi I}\left(\frac{2\sqrt{2}\pi B_s^2 R^6 G^3 M^3 \dot{M}^6}{\mu_0}\right)^{1/7}. \tag{18.41}$$

Example 18.6.3. The X-ray pulsar Centaurus X-3 has a period of 4.84 s and an X-ray luminosity of about $L_x = 5 \times 10^{30}$ W. Assuming that it is a 1.4 $M_\odot$ neutron star with a radius of 10 km, its moment of inertia (assuming for simplicity that it is a uniform sphere) is

$$I = \frac{2}{5} M R^2 = 1.11 \times 10^{38} \text{ kg m}^2.$$

Using Eq. (18.24) for the accretion luminosity, we find the mass transfer rate to be

$$\dot{M} = \frac{R L_x}{GM} = 2.69 \times 10^{14} \text{ kg s}^{-1},$$

or 4.27×10^{-9} $M_\odot$ yr^{-1}. Then, for an assumed magnetic field of $B_s = 10^8$ T and $\alpha = 0.5$, Eq. (18.41) gives the fractional change in the period per second and per year:

$$\frac{\dot{P}}{P} = -2.74 \times 10^{-11} \text{ s}^{-1} = -8.64 \times 10^{-4} \text{ yr}^{-1}.$$

That is, the characteristic time for the period to change is $P/\dot{P} = 1160$ years.

The measured value for Cen X-3 is $\dot{P}/P = -2.8 \times 10^{-4}$ yr^{-1}, smaller than our estimate by a factor of 3 but in good agreement with this simple argument. You may verify that if a 0.85 $M_\odot$ white dwarf with a radius of 6.6×10^6 m and $B_s = 1000$ T is used for the accreting star, rather than a neutron star, then $\dot{P}/P = -1.03 \times 10^{-5}$ yr^{-1}. The measured value is larger by a factor of 27. A white dwarf is hundreds of times larger than a neutron star, so it has a much larger moment of inertia and is more difficult to spin up. The substantially better agreement between the neutron star model and the observations obtained for these systems is compelling evidence that neutron stars are the accreting objects in binary X-ray pulsars.

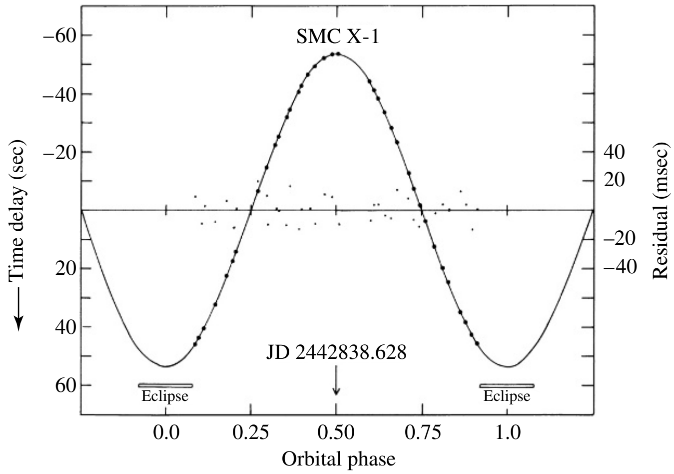

FIGURE 18.24 Measured pulse arrival times (dots) for the binary X-ray pulsar SMC X-1 as a function of its orbital phase. The curve is for the best-fit circular orbit, and the dots about the straight line show the residuals from the best-fit orbit. (Figure adapted from Primini et al., *Ap. J.*, *217*, 543, 1977.)

As an X-ray pulsar orbits its binary companion, the distance from the pulsar to Earth constantly changes. This results in a cyclic variation in the measured pulse period that is analogous to the Doppler shift of a spectral line observed for a spectroscopic binary (see Section 7.3). Figure 18.24 shows the shift in pulse arrival times as a function of the orbital phase for the X-ray pulsar SMC X-1 in the Small Magellanic Cloud. The orbit for this system is almost perfectly circular, with a radius of 53.5 light-seconds = 0.107 AU, less than one-third the size of Mercury's orbit around the Sun.

A complete description of the binary system has been obtained for a small number of *eclipsing* X-ray pulsars with visible companions. Such systems are analogous to double-line, eclipsing, spectroscopic binaries. For example, in the SMC X-1 system the mass of the secondary star is 17.0 $M_\odot$ (with an uncertainty of about 4 $M_\odot$), and its radius is 16.5 $R_\odot$ ($\pm$ 4 $R_\odot$). The masses of the neutron stars have also been determined for these systems. The results are consistent with a neutron star mass of 1.4 $M_\odot$ ($\pm$ 0.2 $M_\odot$), in good agreement with the Chandrasekhar limit.

X-Ray Bursters

If the magnetic field of the neutron star is too weak ($\ll 10^8$ T) to completely disrupt the accretion disk and funnel the accreting matter onto its magnetic poles, these gases will settle over the surface of the star. Without an accretion column to produce a hot spot, X-ray pulses cannot be produced by the rotation of the neutron star. Instead, calculations indicate that when a layer of hydrogen a few meters thick accumulates on the surface, a shell of hydrogen slowly begins burning about a meter below the surface, with a shell of helium burning ignited another meter below that; see Fig. 18.25.[30] This fusion of helium is explosive and releases

[30]This mechanism is reminiscent of the helium shell flashes discussed in Section 13.2.

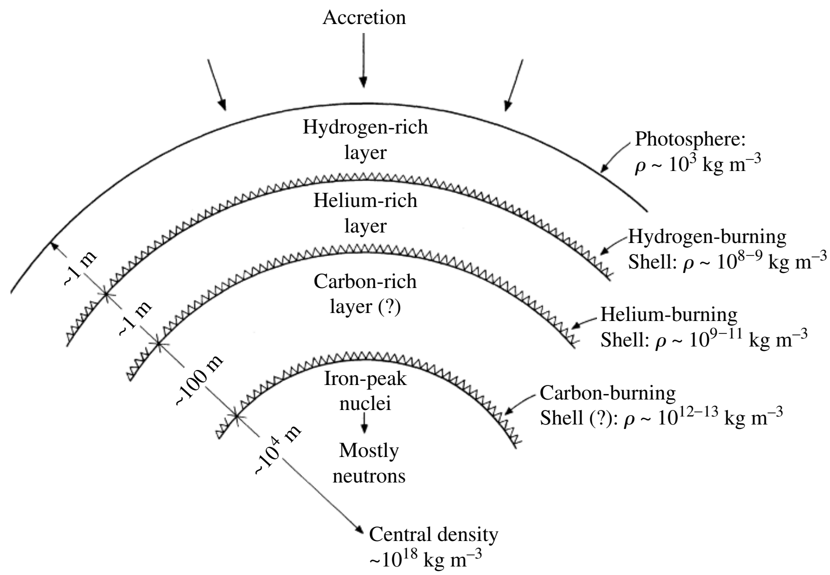

FIGURE 18.25 Surface layers on an accreting neutron star. (Figure adapted from Joss, *Comments Astrophys.*, *8*, 109, 1979.)

a total of $\sim 10^{32}$ J in just a few seconds, with the surface reaching a temperature of about 3×10^7 K (twice the Sun's central temperature). The resulting blackbody spectrum peaks at X-ray wavelengths, and a flood of X-rays is liberated by this **X-ray burster**. Some of the X-rays may be absorbed by the accretion disk and re-emitted as visible light, so an optical flash is sometimes seen a few seconds after the X-ray burst. As the burst luminosity declines in a matter of seconds, the spectrum matches that of a cooling blackbody with a radius of ~ 10 km, consistent with the presence of a neutron star. After a time that can vary from a few hours to a day or more, another layer of hydrogen accumulates and another X-ray burst is triggered.[31] More than 50 X-ray bursters have been found so far. Most are concentrated near the Galactic plane, toward the center of our Galaxy, with some 20% located in old globular clusters.

Low-Mass and Massive X-Ray Binaries

From these and other results, astronomers have identified two classes of binary X-ray systems. The more common type consists of those with low-mass secondary stars (late spectral-type stars with $M_2 \leq 2\,\mathrm{M}_\odot$). These systems belong to the **low-mass X-ray binaries** (LMXBs). LMXBs produce X-ray bursts rather than pulses, indicating that the neutron star's magnetic field is relatively weak. Because low-mass stars are small, the two stars must orbit more closely if mass is to be transferred from one star to the other. For this reason, the LMXBs have short orbital periods, from 33.5 days down to 11.4 minutes. About

[31] It is thought that the gases accreting on X-ray pulsars are constantly undergoing fusion. However, recall from Example 18.1.2 that the energy released in the accretion column will be about 30 times larger, so the energy from fusion will be lost in the glare of the accretion energy.

one-quarter of these systems are found within globular clusters, where the high number density of stars makes the gravitational capture of a neutron star more likely. The neutron stars in LMXBs may also have been formed by the accretion-induced collapse of a white dwarf.

Systems with higher-mass secondaries are referred to as **massive X-ray binaries** (MXRBs). About half of the approximately 130 known MXRBs are X-ray pulsars. With giant or supergiant O and B stars available to fill their Roche lobes, the separation of the stars can be larger and the orbital periods correspondingly longer, from 0.2 days up to 580 days. Even if the secondary star's envelope does not overflow its Roche lobe, the vigorous stellar winds of these stars may still provide the mass transfer rate needed to sustain the production of X-rays. The MXRBs are found near the plane of our Galaxy, where there are young massive stars and ongoing star formation. This is consistent with the idea that an MXRB is the product of the normal evolution of a binary system containing a massive star that survived the supernova explosion of its companion.

So far, only neutron stars have been considered as the accreting object in binary X-ray systems. However, the gravitational potential well is even deeper for matter falling toward a black hole. In this case, up to about 30% of the rest energy of the falling disk material may be emitted as X-rays. In fact, as was discussed in Section 17.3, these systems provide the best evidence for the existence of stellar-mass black holes. The gas spilling through the inner Lagrangian point is heated to millions of kelvins as it spirals down through the black hole's accretion disk (see Fig. 17.24) and so emits X-rays. The identification of a black hole rests on determining that the mass of a compact, X-ray-emitting object exceeds the approximately 3 $M_\odot$ upper limit for the mass of a rapidly rotating neutron star. Thus the procedure for detecting a black hole in a binary X-ray system is similar to that used to measure the masses of neutron stars in these systems.

At present, there are only a handful of X-ray binaries that allow such a dynamical determination of the masses involved. The best cases at the time of this writing are A0620−00, V404 Cygni, Cygnus X-1, and LMC X-3. Since none of these systems exhibit eclipses, the resulting uncertainty about their orbital inclinations means that the masses calculated are lower limits (see Section 7.3). A0620−00 is an **X-ray nova**, powered by the sporadic accretion of material from its companion, a K5 main-sequence star. The relative faintness of the secondary star allows the measurement of the radial velocity of *both* the accretion disk and the companion star. The identification of A0620−00 as a 3.82 ± 0.24 $M_\odot$ black hole seems secure, as was described in Section 17.3 and Problem 17.24. V404 Cyg is also an X-ray nova, where recent measurements persuasively document the presence of a 12 $M_\odot$ black hole (Section 17.3).

The arguments for the other two systems, although strong, are not as conclusive. Neither has a fully developed accretion disk, and so the velocities of both members cannot be determined. Cygnus X-1, perhaps the best-known black hole candidate, is a bright MXRB. Because almost all of the light comes from the secondary, Cyg X-1 is essentially a single-line spectroscopic binary. The identification of Cyg X-1 as a black hole therefore depends on the identification of the secondary star (HDE 226868) as a O9.7 Iab supergiant with a mass of 17.8 $M_\odot$. The most likely result, making reasonable assumptions about this binary system, is that the mass of the compact object in Cyg X-1 is 10.1 $M_\odot$. Even the worst-case argument results in a secure lower limit of 3.4 $M_\odot$, providing the evidence that Cyg X-1 is a black hole.

The secondary star in the LMC X-3 system is a B3 main-sequence star that is orbiting an unseen, more massive companion. Although the lower limit on the mass of the compact companion is 3 $M_\odot$, a more probable mass range is 4–9 $M_\odot$—again, solid evidence for a black hole. Other X-ray binary systems may contain black holes, such as Nova Mus 1991 in the southern constellation Musca (the Fly), LMC X-3 in the Large Magellanic Cloud, and CAL 87 (in the direction of the Large Magellanic Cloud), but the evidence in these cases is not yet as strong.

SS 433

One more X-ray binary and possible black hole candidate should be mentioned: SS 433, one of the most bizarre objects known to astronomers.[32] In 1978, it was discovered that this object displays *three* sets of emission lines. One set of spectral lines was greatly blueshifted, another set was greatly redshifted, and a third set lacked a significant Doppler shift. Here was an object with three components: Two were approaching and receding, respectively, at one-quarter the speed of light while the third stayed nearly still! The wavelengths of the shifted lines vary with a period of 164 days, while the wavelengths of the nearly stationary lines show a smaller shift with a 13.1-day period. Furthermore, the position of SS 433 lies at the center of a diffuse, elongated shell of gas known as W50, which is probably a supernova remnant.

The 13.1-day period of SS 433 describes the orbit of a compact object (most probably a neutron star, but perhaps a black hole) around the primary. The primary is thought to be a 10–20 $M_\odot$ early-type star with a stellar wind that produces the broad stationary emission lines.[33] Surrounding the compact object is an accretion disk that contributes to the visible light from the system equally with the secondary. A tidal interaction between the disk and the two stars could be responsible for a precessional wobble of the disk that has a period of 164 days, analogous to Earth's 25,770-year precessional wobble discussed in Section 1.3. There is broad agreement that the varying Doppler-shifted emission lines, shown in Fig. 18.26, come from two *relativistic jets* that expel particles at $0.26c$ in opposite directions along the axis of the disk. The jets are probably powered by the accretion of matter at a rate exceeding the Eddington limit, generating X-rays at a prodigious rate. This could produce a radiation pressure sufficient to expel a portion of the accreting gases at relativistic speeds in the direction of least resistance—perpendicular to the disk. As the disk precesses, two oppositely directed jets sweep out a cone in space every 164 days, resulting in cyclic variations in both the radial velocity of the jets and the observed Doppler shift. The collimation of the jets could be the result of the ionized gases moving along magnetic field lines. The axis of the precessional cone makes an angle of 79° with the line of sight; the cone's axis is also closely aligned with the long axis of the probable supernova remnant, W50. In fact, there are two regions that have been observed to emit X-rays, presumably

[32]"SS" stands for the catalog of peculiar emission-line stars compiled by Bruce Stephenson and Nicholas Sanduleak.

[33]For example, one recent measurement of SS 433 favors a 0.8 $M_\odot$ neutron star orbiting a 3.2 $M_\odot$ companion. Some astronomers have suggested that the primary may be a Wolf–Rayet star (Section 15.3) to account for the broad stationary emission lines. Although a substantial percentage of Wolf–Rayet stars are found in binaries, these stars' own energetic winds, rather than the transfer of mass in a close binary system, seem to be responsible for removing most of their hydrogen envelopes.

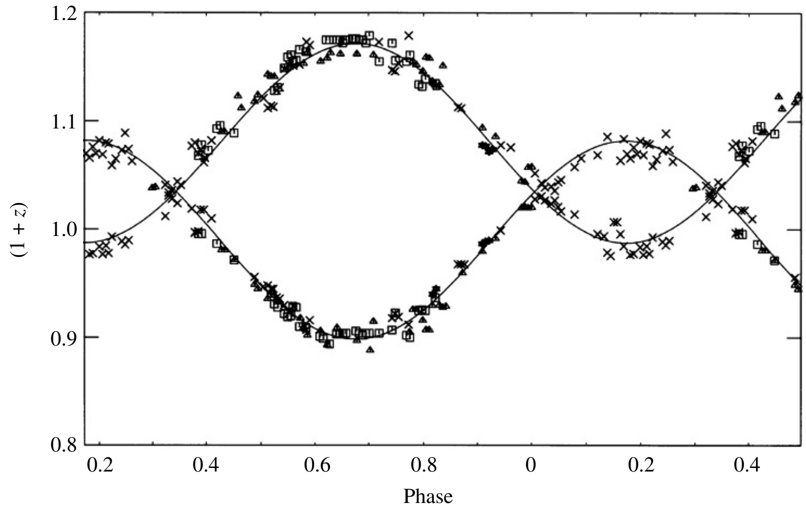

FIGURE 18.26 Doppler shifts measured for the emission lines in SS 433. From Eq. (4.38), $z = 0.1$ and $z = 0.2$ correspond to speeds of 28,500 km s^{-1} and 54,100 km s^{-1}, respectively. (Figure adapted from Margon, Grandi, and Downes, *Ap. J.*, *241*, 306, 1980.)

where the jets collide with the remnant's gases and heat them to about 10^7 K. Figure 18.27 shows the general features of this incredible system.

The Fate of Binary X-Ray Systems

What is the fate of a binary X-ray system? As it reaches the endpoint of its evolution, the secondary star will end up as a white dwarf, neutron star, or black hole. The effect on the system depends on the mass of the secondary star. In low-mass systems (LMXBs), the companion star will become a white dwarf without disturbing the circular orbit of the system. On the other hand, the higher-mass secondary in a MXRB may explode as a supernova. If more than half of the system's mass is retained (Eq. 18.35), a pair of neutron stars will circle each other in orbits that probably have been elongated by the blast. Otherwise, the supernova may disrupt the system and hurl the solitary neutron stars into space. This is consistent with observations that pulsars (like MXRBs) are concentrated near the plane of our Galaxy and may have high space velocities that can exceed 1000 km s^{-1}.

Millisecond Radio Pulsars

The principal way in which a binary system containing two neutron stars can be detected is if at least one of them is a pulsar. Astronomers therefore search for cyclic variations in the measured periods of radio pulsars, analogous to the effect described here for the X-ray pulsars. Although half of all stars in the sky are actually multiple systems, *none* of the first one hundred pulsars discovered belonged to a binary. The first binary pulsar, PSR 1913+16, was discovered in 1974 by American astronomers Russell Hulse and Joseph Taylor, using the Arecibo radio telescope. The search strategy for binary pulsars changed with the 1982

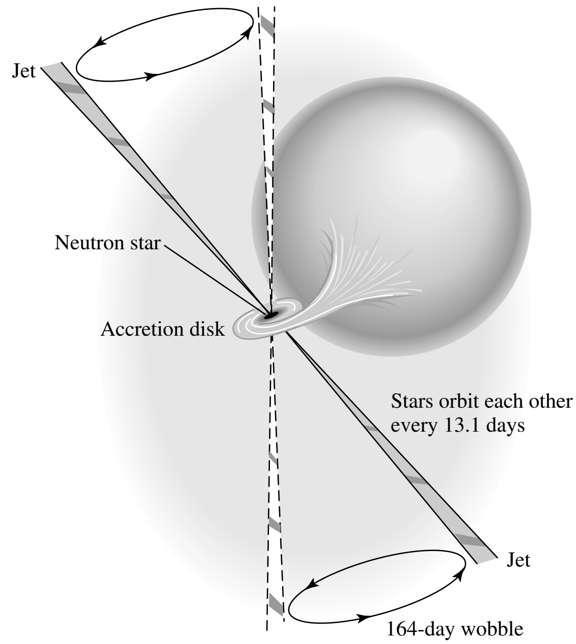

FIGURE 18.27 SS 433. The axis of the cone swept out by the precessing jets makes a 79° angle with the line of sight.

discovery by Donald Backer of UC Berkeley and his colleagues of the then-fastest known pulsar, PSR 1937+214. With a period of 1.558 milliseconds, this pulsar spins 642 times each second.[34] Although this astounding rotation rate seemed to indicate a young pulsar, the very small value of the period derivative ($\dot{P} = 1.051054 \times 10^{-19}$) implies a weak magnetic field ($\approx 8.6 \times 10^4$ T; see Example 16.7.2) and a very old pulsar. From Problem 16.22, the age of the pulsar may be estimated as $P/2\dot{P} = 235$ million years, an order of magnitude older than previously discovered pulsars.[35] Although PSR 1937+214 is an isolated pulsar, the paradox of the oldest pulsar also being among the fastest quickly brought astronomers to a surprising conclusion: PSR 1937+214 must once have been a member of a low-mass X-ray binary system. (Recall that, like PSR 1937+214, LMXBs have weak magnetic fields.) Accretion from the secondary star could have spun up the neutron star to its present rapid rate. The neutron star's magnetic field may also have been rejuvenated by this process, although the details of how this might occur are not yet clear.

A likely evolutionary picture has emerged that brings the observations of binary X-ray sources and binary pulsars together. In this scenario, there are two classes of binary pulsars. Those with high-mass companions (neutron stars) have shorter periods and eccentric orbits and probably result from the evolution of a massive X-ray binary system. (Recall that an

[34] Middle C on a piano has an audible frequency of 262 Hz. The pulsar's rotation frequency is more than an octave higher, between D# and E!

[35] $P/2\dot{P}$ is an estimate of a pulsar's age only if the pulsar's spin has not been affected by accretion.

MXRB that managed to retain more than half of its mass following the supernova of the companion star would produce such a pair of neutron stars with elongated orbits.) The other class of binary pulsars are characterized by low-mass companions (white dwarfs), longer orbital periods, and circular orbits. These are probably the descendants of low-mass X-ray binary systems.

Because LMXBs are common in globular clusters, radio astronomers slued their telescopes toward these targets and discovered more binary and millisecond pulsars (those with periods less than approximately 10 ms). Numerous surveys, including one conducted by the Chandra X-Ray Observatory, suggest that 47 Tuc may have more than 300 neutron stars, approximately 25 of which are millisecond pulsars. The mounting statistics make it clear that most of the globular cluster pulsars are members of binaries and that most (but not all) are millisecond pulsars. (Conversely, most of the known millisecond pulsars have been found in globular clusters.) If these pulsars are the evolutionary product of LMXBs, then how can the absence of a white dwarf companion be explained for a significant minority of them?

Black Widow Pulsars

An answer may be found from observations of PSR 1957+20. It is a rarity: a binary millisecond pulsar that eclipses its companion, a meager 0.025 $M_\odot$ white dwarf. However, the eclipses last for some 10% of the orbit, implying that the light is blocked by an object larger than the Sun. Significantly, the dispersion of the pulsar signal (see page 595) increases just before and after the eclipse, indicating that the white dwarf is surrounded by ionized gas. The pulsar seems to be *evaporating* its white dwarf companion with its energetic beam of photons and charged particles. Within a few million years, the white dwarf may disappear, devoured by this **black widow pulsar**; see Fig. 18.28.

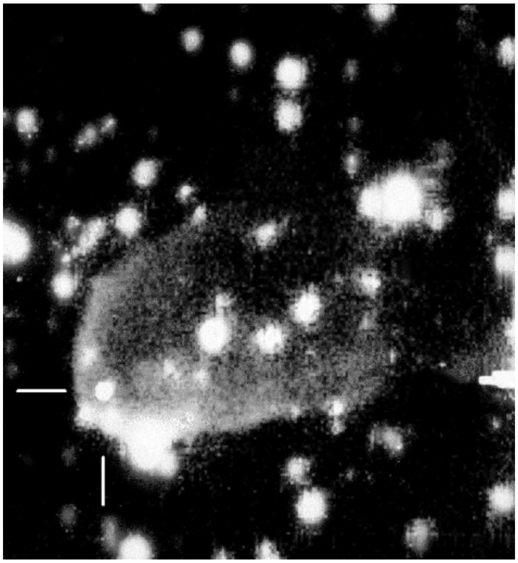

FIGURE 18.28 Gas being removed by the "black widow pulsar," PSR 1957+20. The pulsar is at the intersection of the white lines. (Photo courtesy of S. Kulkarni and J. Hester, Caltech.)

Another example of the ablation of an eclipsing millisecond pulsar's companion has been found for PSR 1744−24A in the globular cluster Terzan 5, where the eclipses last for half of the orbital period. It is possible that some of the evaporated material may form a disk of gas and dust around the pulsar that could eventually (after a million years or so) condense and form planets around the pulsar. Or, if the evaporation of the companion star is incomplete, a planet-size remnant could be left orbiting the pulsar.

Mechanisms such as these may be responsible for the three planets thought to be traveling in circular orbits around PSR 1257+12, some 500 pc away in the constellation Virgo. As determined from a careful analysis of pulse arrival times, the innermost planet has a mass of 0.015 $M_\oplus$ that is 0.19 AU from the pulsar, followed by a 3.4 $M_\oplus$ object that is at a distance of 0.36 AU. The outermost planet's mass is 2.8 $M_\oplus$, and it is at a distance of 0.47 AU.

As more millisecond pulsars are discovered, it should become clear whether the foregoing evolutionary picture is correct.

Double Neutron Star Binaries

A small number of detached binary systems are known to exist in which both both members are neutron stars. As highly relativistic systems with no current mass exchange between the system members, these **double neutron star binaries** are exquisite natural laboratories for the testing of predictions of the General Theory of Relativity.

The first such system discovered is the **Hulse–Taylor pulsar**, PSR 1913+16, with an orbital separation just a little larger than the Sun's diameter. A 30-year study of this system has confirmed the existence of gravitational waves.[36]

Nearly everything is known about the Hulse–Taylor system with incredible precision, as can be seen by inspecting the observational data in Table 18.1. Because of this level of precision, this binary system provides an ideal natural laboratory for testing Einstein's theory of gravity. For example, recall from Section 17.1 that as Mercury passes through the

TABLE 18.1 Data for the Hulse–Taylor Pulsar, PSR 1913+16.

Parameter	Value	Uncertainty
Pulse Frequency (ω)	16.94053918425292 Hz	$\pm\ 15 \times 10^{-14}$ Hz
Pulse Frequency Derivative ($\dot{\omega}$)	-2.47583×10^{-15} Hz s^{-1}	$\pm\ 3 \times 10^{-20}$ Hz s^{-1}
Mass (pulsar)	1.4414 $M_\odot$	$\pm$ 0.0002 $M_\odot$
Mass (companion)	1.3867 $M_\odot$	$\pm$ 0.0002 $M_\odot$
Eccentricity (e)	0.6171338	$\pm$ 0.0000004
Period of Orbit ($P_{\rm orb}$)	0.322997448930 d	$\pm\ 4 \times 10^{-13}$ d
Period of Oribt Derivative ($\dot{P}_{\rm orb}$)	-2.4056×10^{-12}	$\pm\ 0.0051 \times 10^{-12}$
Periastron shift ($\dot{\omega}_{\rm orb}$)	4.226595° yr^{-1}	$\pm$ 0.000005° yr^{-1}

Notes:

1. Data for ω and $\dot{\omega}$ from epoch January 14, 1986; see http://www.atnf.csiro.au/research/pulsar/psrcat/

2. Remaining data from J. M. Weisberg and J. H. Taylor (2005).

[36]Russell Hulse and Joseph Taylor shared the 1993 Nobel Prize for their discovery of PSR 1913+16.

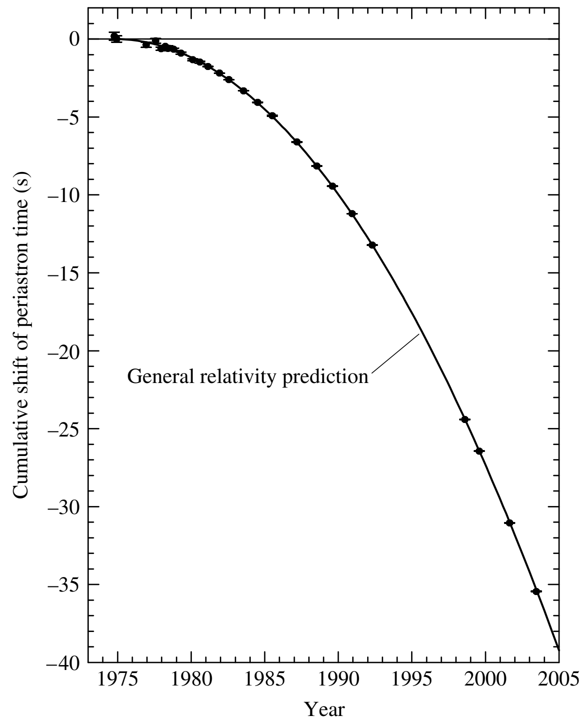

FIGURE 18.29 Observations (dots) of the delay in the time of periastron for PSR 1913+16, compared with the prediction of the theory of general relativity (solid line). [Adapted from a figure courtesy of J. M. Weisberg and J. H. Taylor (2005).]

curved spacetime near the Sun, the position of perihelion in its orbit is shifted by 43″ per century; see Fig. 17.1. For PSR 1913+16, general relativity predicts a similar shift in the point of periastron, where the two neutron stars are nearest each other. The theoretical value is in excellent agreement with the measurement of $4.226595 \pm 0.000005^\circ\ \mathrm{yr}^{-1}$ (35,000 times Mercury's rate of shift). This effect on the orbit is cumulative; with every orbit, the pulsar arrives later and later at the point of periastron. Figure 18.29 shows the incredible agreement between theoretical and observed values of the accumulating time delay.

The most spectacular aspect of the studies of PSR 1913+16 is the confirmation of the existence of gravitational radiation. As the two neutron stars move in their orbits, gravitational waves carry energy away from the system and the orbital period decreases. According to general relativity, the rate at which the orbital period changes as a consequence of the emission of gravitational quadrupole radiation[37] is

$$\dot{P}_{\mathrm{orb}} = \frac{dP_{\mathrm{orb}}}{dt} = -\frac{96}{5}\frac{G^3M^2\mu}{c^5}\left(\frac{4\pi^2}{GM}\right)^{4/3}\frac{f(e)}{P_{\mathrm{orb}}^{5/3}}, \tag{18.42}$$

[37]The term *quadrupole* describes the geometry of the emitted gravitational radiation, just as electric dipole radiation describes the electromagnetic radiation emitted by two electric charges moving around each other.

where

$$M = M_1 + M_2$$

$$\mu = \frac{M_1 M_2}{M_1 + M_2}$$

and $f(e)$ describes the effect of the eccentricity of the orbit,

$$f(e) = \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right)\left(1 - e^2\right)^{-7/2}.$$

(There are also higher-order correction terms that have been neglected here.) Inserting the preceding values for the masses and eccentricity, the theoretical rate of orbital period decay is calculated to be $\dot{P}_{\rm orb,predict} = -(2.40242 \pm 0.00002) \times 10^{-12}$, which agrees with the measured value of $\dot{P}_{\rm orb,meas} = -(2.4056 \pm 0.0051) \times 10^{-12}$ to within 0.13%. In presenting the results of an earlier calculation of the orbital period decay in 1984, Joel Weisberg and Joseph Taylor wrote, "It now seems an inescapable conclusion that gravitational radiation exists as predicted by the general relativistic quadrupole formula." Astronomers are fortunate to have caught this superb natural laboratory before it disappears. As the separation of the neutron stars shrinks by about 3 mm per orbit, the system will coalesce some 300 million years in the future.

Another tremendous natural laboratory for testing general relativity was discovered in 2003. This double neutron star system is actually a binary pulsar system. J0737–3039A has a pulse period of $P_A = 0.02269937855615 \pm 6 \times 10^{-14}$ s, and J0737–3039B has a pulse period of $P_B = 2.7734607474 \pm 4 \times 10^{-10}$ s. As with the Hulse–Taylor pulsar, this (thus far) unique system provides a valuable test of orbital precession and gravitational radiation. However, J0737–3039A/B can also test the prediction of delayed arrival times for signals from one pulsar passing through the gravitational well of the other pulsar (recall Fig. 17.4). As their signals interact with each other's magnetic field and with the plasma environment, they provide an opportunity to test theories about plasma physics as well. It may also be possible to measure the moments of inertia of the pulsars, providing important tests of the interior structure models of neutron stars, including their exotic equations of state.

Short–Hard Gamma Ray Bursts

What will be the consequence of the merger of the two neutron stars? In Section 15.4 two classes of gamma ray bursts were discussed. Extensive observations have confirmed that long–soft gamma ray bursts (> 2 s) are extreme examples of core-collapse supernovae (collapsars or supranovas). On the other hand, it is now believed that **short–hard gamma ray bursts** (< 2 s) are the result of the mergers of compact objects, either two neutron stars or a neutron star and a black hole.

The first clear detections of mergers of compact objects in binaries were obtained by the Swift and HETE-2 spacecraft in 2005. The July 9, 2005, event in particular also produced a visible-light afterglow that allowed astronomers to unambiguously identify the host galaxy. Short–hard gamma ray bursts emit about 1000 times less energy than the long–soft events do.

The Hulse–Taylor system is destined to produce a short–hard gamma ray burst, although it may or may not be observable from Earth, depending on the orientation of the jet.

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PROBLEMS

18.1 Use the ideal gas law to argue that in a close binary system, the temperature of a star's photosphere is approximately constant along an equipotential surface. What effect could the proximity of the other star have on your argument?

18.2 Each of the Lagrange points L_4 and L_5 forms an equilateral triangle with masses M_1 and M_2 in Fig. 18.3. Use this to confirm the value of the effective gravitational potential at L_4 and L_5 given in the figure caption.

18.3 **(a)** Consider a gas of density ρ moving with velocity v across an area A perpendicular to the flow of the gas. Show that the rate at which mass crosses the area is given by Eq. (18.11).

(b) Derive Eq. (18.12) for the radius of the intersection of two identical overlapping spheres, when $d \ll R$.

18.4 Use Eq. (18.19) to show that the maximum disk temperature is found at $r = (49/36)R$ and is equal to $T_{\text{max}} = 0.488T_{\text{disk}}$.

18.5 Integrate Eq. (18.15) for the ring luminosity from $r = R$ to $r = \infty$ [with Eq. (18.19) for the disk temperature]. Does your answer agree with Eq. (18.23) for the disk luminosity?

18.6 Consider an "average" dwarf nova that has a mass transfer rate of

$$\dot{M} = 10^{13.5}\ \mathrm{kg\ s^{-1}} = 5 \times 10^{-10}\ \mathrm{M_\odot\ yr^{-1}}$$

during an outburst that lasts for 10 days. Estimate the total energy released and the absolute magnitude of the dwarf nova during the outburst. Use values for Z Cha's white dwarf from Example 18.4.1. Neglect the small amount of light contributed by the primary and secondary stars.

18.7 Assume that the absolute bolometric magnitude of a dwarf nova during quiescence is 7.5 and that it brightens by three magnitudes during outburst. Using values for Z Cha, estimate the rate of mass transfer through the accretion disk.

18.8 When the accretion disk in a cataclysmic variable is eclipsed by the secondary star, the blueshifted emission line is the first to disappear at the beginning of the eclipse, and the redshifted emission line is the last to reappear when the eclipse ends. What does this have to say about the directions of rotation of the binary system and the accretion disk?

18.9 **(a)** Show that in a close binary system where angular momentum is conserved, the change in orbital period produced by mass transfer is given by

$$\frac{1}{P}\frac{dP}{dt} = 3\dot{M}_1 \frac{M_1 - M_2}{M_1 M_2}.$$

(b) U Cephei (an Algol system) has an orbital period of 2.49 days that has increased by about 20 s in the past 100 years. The masses of the two stars are $M_1 = 2.9\ \mathrm{M_\odot}$ and $M_2 = 1.4\ \mathrm{M_\odot}$. Assuming that this change is due to the transfer of mass between the two stars in this Algol system, estimate the mass transfer rate. Which of these stars is gaining mass?

18.10 Algol (the "demon" star, in Arabic) is a semidetached binary. Every 2.87 days, its brilliance is reduced by more than half as it undergoes a deep eclipse, its apparent magnitude dimming from 2.1 to 3.4. The system consists of a B8 main-sequence star and a late-type (G or K) subgiant; the deep eclipses occur when the larger, cooler star (the subgiant) moves in front of its smaller, brighter companion. The "Algol paradox," which troubled astronomers in the first half of the twentieth century, is that according to the ideas of stellar evolution discussed in Section 10.6, the more massive B8 star should have been the first to evolve off the main sequence. What is your solution to this paradox? (The Algol system actually contains a third star that orbits the other two every 1.86 years, but this has nothing to do with the solution to the Algol paradox.)

Algol may be easily found in the constellation Perseus (the Hero, who rescued Andromeda in Greek mythology).

18.11 Consider a $10^{-4}\ \mathrm{M_\odot}$ layer of hydrogen on the surface of a white dwarf. If this layer were completely fused into helium, how long would the resulting nova last (assuming a luminosity equal to the Eddington luminosity)? What does this say about the amount of hydrogen that actually undergoes fusion during a nova outburst?

18.12 Consider a layer of $10^{-4}\ \mathrm{M_\odot}$ of hydrogen on the surface of a white dwarf. Compare the gravitational binding energy before the nova outburst to the kinetic energy of the ejected layer when it has traveled far from the white dwarf and has a speed of 1000 km $\mathrm{s^{-1}}$.

18.13 In this problem, you will examine the fireball expansion phase of a nova shell. Suppose that mass is ejected by a nova at a constant rate of $\dot{M}_{\mathrm{eject}}$ and at a constant speed v.

(a) Show that the density of the expanding shell at a distance r is

$$\rho = \dot{M}_{\rm eject}/4\pi r^2 v.$$

(b) Let the mean opacity, $\overline{\kappa}$, of the expanding gases be a constant. Suppose that at some time $t = 0$, the outer radius of the shell was R, and the radius of the photosphere, where $\tau = 2/3$, was R_0. Show that

$$\frac{1}{R} = \frac{1}{R_0} - \frac{1}{R_\infty},$$

where

$$R_\infty \equiv \frac{3\overline{\kappa}\dot{M}_{\rm eject}}{8\pi v}.$$

(The reason for the "∞" subscript will soon become clear.)

(c) At some later time t, the radius of the shell will be $R + vt$ and the radius of the photosphere will be $R(t)$. Show that

$$\frac{1}{R + vt} = \frac{1}{R(t)} - \frac{1}{R_\infty}.$$

(d) Combine the results from parts (b) and (c) to write

$$R(t) = R_0 + \frac{vt(1 - R_0/R_\infty)^2}{1 + (vt/R_\infty)(1 - R_0/R_\infty)},$$

(e) Argue that terms containing R_0/R_∞ are very small and can be ignored, and so obtain

$$R(t) \simeq \frac{vt}{1 + vt/R_\infty}.$$

(f) Show that the fireball's photosphere initially expands linearly with time and then approaches the limiting value of R_∞, in agreement with Eq. (18.30).

(g) Using the data given in the text following Eq. (18.31), make a graph of $R(t)$ vs. t for the five days after the nova explodes. The "knee" in the graphs marks the end of the linear expansion period; estimate when this occurs. How does this compare with the duration of the optically thick fireball phase of the nova?

18.14 Use Eq. (18.31) to estimate the photospheric temperature of a nova fireball, adopting the Eddington luminosity for the luminosity of the fireball.

18.15 Assuming that the hydrostatic-burning phase of a nova lasts for 100 days, find the (constant) rate at which mass is ejected, $\dot{M}_{\rm eject}$, for a surface layer of 10^{-4} $M_\odot$.

18.16 For each kilogram of a carbon–oxygen composition (30% ${}^{12}_{6}C$) that is burned to produce iron, 7.3×10^{13} J of energy is released. Assuming an initial 1.38 $M_\odot$ white dwarf with a radius of 1600 km, how much iron would have to be produced to cause the star to be gravitationally unbound? How much additional iron would have to be manufactured to produce a Type Ia supernova with an average ejecta speed of 5000 km s^{-1}? Take the gravitational potential energy to be -5.1×10^{43} J for a realistic white dwarf model, and express your answers in units of $M_\odot$.

18.17 Use Eqs. (7.4), (18.33), and (18.34) to derive Eq. (18.35), the condition for a supernova to disrupt a binary system.

18.18 **(a)** Show that the Alfvèn radius is given by Eq. (18.39).

(b) Show that $\dot{P}/P$ for the spin-up of an X-ray pulsar is given by Eq. (18.41).

18.19 Find the value of the magnetic field for which the Alfvèn radius is equal to the radius of the white dwarf found in Example 18.2.1. Do the same thing for the neutron star used in that example.

18.20 Estimate the lifetime of a binary X-ray system using the information in Example 18.2.1. Take the lifetime to be the time required to transfer a mass of 1 $M_\odot$.

18.21 The X-ray pulsar 4U0115+63 has a period of 3.61 s and an X-ray luminosity of about $L_x = 3.8 \times 10^{29}$ W. Assuming that it is a 1.4 $M_\odot$ neutron star with a radius of 10 km and a surface magnetic field of 10^8 T, find its mass transfer rate, $\dot{M}$, and the value of $\dot{P}/P$. Repeat these calculations assuming that this object is a 0.85 $M_\odot$ white dwarf with a radius of 6.6×10^6 m and a surface magnetic field of 1000 T. For which of these models do you obtain better agreement with the measured value of $\dot{P}/P = -3.2 \times 10^{-5}$ yr^{-1}?

18.22 **(a)** Use Eq. (18.24) to show that the spin-up rate can be written as

$$\log_{10}\left(-\frac{\dot{P}}{P}\right) = \log_{10}\left(P L_{\text{acc}}^{6/7}\right) + \log_{10}\left[\frac{\sqrt{\alpha}}{2\pi I}\left(\frac{2\sqrt{2}\pi B_s^2 R^{12}}{\mu_0 G^3 M^3}\right)^{1/7}\right].$$

The term on the left and the first term on the right consist of quantities that can be measured observationally. The second term on the right depends on the specific model (neutron star or white dwarf) of the X-ray pulsar.

(b) Make a graph of $\log_{10}(-\dot{P}/P)$ (vertical axis) vs. $\log_{10}\left(P L_{\text{acc}}^{6/7}\right)$ (horizontal axis). Use the values from Example 18.6.3 to plot two lines, one for a neutron star and one for a white dwarf. Let $\log_{10}\left(P L_{\text{acc}}^{6/7}\right)$ run from 25 to 29.

(c) Use the data in Table 18.2 to plot the positions of six binary X-ray pulsars on your graph. (You will have to convert $-\dot{P}/P$ into units of s^{-1}.)

(d) Which model of a binary X-ray pulsar is in better agreement with the data? Comment on the position of Her X-1 on your graph.

TABLE 18.2 X-ray Pulsar Data for Problem 18.22. (Data from Rappaport and Joss, *Nature*, *266*, 683, 1977, and Joss and Rappaport, *Annu. Rev. Astron. Astrophys.*, *22*, 537, 1984.)

System	P (s)	L_{acc} (10^{30} W)	$-\dot{P}/P$ (yr^{-1})
SMC X-1	0.714	50	7.1×10^{-4}
Her X-1	1.24	1	2.9×10^{-6}
Cen X-3	4.84	5	2.8×10^{-4}
A0535+26	104	6	3.5×10^{-2}
GX301−2	696	0.3	7.0×10^{-3}
4U0352+30	835	0.0004	1.8×10^{-4}

18.23 **(a)** Consider an X-ray burster that releases 10^{32} J in 5 seconds. If the shape of its peak spectrum is that of a 2×10^7 K blackbody, estimate the radius of the underlying neutron star.

(b) In Problem 17.9 you showed that using the Stefan–Boltzmann formula to find the radius of a compact blackbody can lead to an overestimate of its radius. Use Eq. (17.33) to find a more accurate value for the radius of the neutron star.

18.24 Make a scale drawing of the SMC X-1 binary pulsar system, including the size of the secondary star. Assuming that the primary is a 1.4 $M_\odot$ neutron star, locate the system's center of mass and its inner Lagrangian point, L_1. (You can omit the accretion disk.)

18.25 The relativistic ($v/c = 0.26$) jets coming from the accretion disk in SS 433 sweep out cones in space as the disk precesses. The central axis of these cones makes an angle of 79° with the line of sight, and the half-angle of each cone is 20°. This means that at some point in the precession cycle, the jets are moving perpendicular to the line of sight. Yet, from Fig. 18.26, the radial velocities obtained from the Doppler-shifted spectral lines do *not* cross at zero radial velocity, but at $\sim$ 10,000 km s^{-1}. Use Eq. (4.32) to explain this discrepancy in terms of a transverse Doppler shift. (You can ignore the speed of the SS 433 binary system itself, which is only about 70 km s^{-1}.)

18.26 The distance to SS 433 is about 5.5 kpc, and the angular separation of SS 433 and the X-ray emitting regions (where the jets interact with the gases of W50) extends as far as 44′. Estimate a lower limit for the amount of time the jets have been active.

18.27 PSR 1953+29 is a millisecond pulsar with a period of 6.133 ms. The measured period derivative for PSR 1953+29 is $\dot{P} = 3 \times 10^{-20}$. Use Problem 16.22 to estimate the age of this millisecond pulsar, assuming that no accretion has occurred to alter the pulsar's spin. Also, use Eq. (16.33) to estimate the value of this pulsar's magnetic field.

18.28 Integrate Eq. (18.41) for the spin-up of an X-ray pulsar to estimate the time for a millisecond pulsar to be spun up to a final period of 1 ms from an initial period of 100 s (longer than the longest known pulsar period of 11.7 s, and within the range of X-ray pulsar periods). Assume a 1.4 $M_\odot$ neutron star with a radius of 10 km. Use $B_s = 10^4$ T for the magnetic field and $\dot{M} = 10^{14}$ kg s^{-1} for the mass transfer rate. How much mass is transferred in that time (in kilograms and in solar masses)?

18.29 The three planets orbiting PSR 1257+12 have orbital periods of 25.34 d, 66.54 d, and 98.22 d. Verify that these objects obey Kepler's third law.

18.30 **(a)** Use Kepler's third law to find the semimajor axis of the orbit of the binary pulsar PSR 1913+16.

(b) What is the change in the semimajor axis after one orbital period of the pulsar?

COMPUTER PROBLEMS

18.32 **(a)** Use the `StatStar` model data on page 283 and Eq. (18.14) to make a graph of $\log_{10} \dot{M}$ (vertical axis) vs. $\log_{10} d$ (horizontal axis). Use the slope of your graph to find how the mass transfer rate, $\dot{M}$, depends on d.

(b) Use Eqs. (L.1) and (L.2) to show that $\dot{M} \propto d^{4.75}$ near the surface, and so verify that the mass transfer rate increases rapidly with the overlap distance d of two stars. Note that your answer to part (a) will be slightly different from this because of the density-dependence of `tog_bf` (the ratio of the guillotine factor to the gaunt factor) calculated in the `Opacity` routine.

18.33 Use Eq. (18.19) and Wien's law to make two log-log graphs: (1) the disk temperature, $T(r)$, and (2) the peak wavelength $[\lambda_{\max}(r)]$ of the blackbody spectrum, for the accretion disk around the black hole A0620−00 as a function of the radial position r. For this system, the mass of the black hole is 3.82 $M_\odot$, the mass of the secondary star 0.36 $M_\odot$, and the period of the orbit is 0.3226 day. Assume $\dot{M} = 10^{14}$ kg s^{-1} (about 10^{-9} $M_\odot$ yr^{-1}), and use the Schwarzschild radius, R_S (Eq. 17.27), for the radius of the black hole. (On your graph, plot r/R_S rather than r.) For a nonrotating black hole, the last stable orbit for a massive particle is at $3R_S$, so use this as the inner edge of the disk. Let the outer edge of the disk be determined by Kepler's third law along with Eqs. (18.25) and (18.26). On your log-log graph of $\lambda_{\max}$ vs. r/R_S, identify the regions of the disk that emit X-ray, ultraviolet, visible, and infrared radiation.

PART III

The Solar System

CHAPTER 19

Physical Processes in the Solar System

19.1 ■ A BRIEF SURVEY

In Chapter 11 we studied the Sun, by far the largest member of the Solar System, in some detail. The question of its formation, along with the formation of other stars, was discussed in Chapter 12. As we have seen, based on the observations of protostars and very young stars (e.g., HH 30, Vega, β Pictoris, and the proplyds in the Orion Nebula; see page 437, along with Figs. 12.19–12.23), it is evident that a natural extension of the formation of many stars includes the accompanying formation of planetary systems that develop within equatorial disks of material that orbit the newborn stars. In fact the first confirmation of an extrasolar planet around a main-sequence star (51 Pegasi) was announced in 1995. After that initial announcement, a total of 155 extrasolar planets were discovered in just the next ten years alone. In Part III we will study one well-known example of a planetary system in some detail, namely our own. We will also consider the growing body of information regarding extrasolar planets. However, it is beyond the scope of this text to describe all of the fascinating details of each of the planets in our Solar System and their moons, not to mention the meteorites, asteroids, comets, Kuiper Belt objects, and interplanetary dust; that is left to the many excellent books dedicated to the subject. Rather, we will consider the basic features of these objects and extrasolar planets in the context of stellar evolution, together with some of the underlying physical processes that have helped to shape them.

General Characteristics of the Planets

The planets have long been studied from Earth, first with the naked eye and later with telescopes. Since the advent of space flight, we have sent manned and unmanned spacecraft to our Moon, and, with the exception of Pluto, we have visited (with unmanned probes) each of the other planets in the Solar System.

Each of the planets (excluding Pluto, 2003 UB313,[1] and other members of the Kuiper belt) can be thought of as belonging to one of two major groups. The rocky **terrestrial**

[1]2003 UB313 was discovered in January 2005, based on images obtained in 2003 (see page 827). As of May 2006, an official classification of 2003 UB313 as a major planet or a minor planet had not yet been made, nor has a formal name been given to the object. These official designations are made by the International Astronomical Union.

TABLE 19.1 General Characteristics of the Planets. The range of values for some features of the terrestrial and giant planets ($M_\oplus$ and $R_\oplus$ represent the mass and radius of Earth, respectively).

Characteristic	Terrestrial	Giant
Basic form	Rock	Gas/Ice/Rock
Mean orbital distance (AU)	0.39–1.52	5.2–30.0
Mean "surface" temperature (K)	215–733	70–165
Mass ($M_\oplus$)	0.055–1.0	14.5–318
Equatorial radius ($R_\oplus$)	0.38–1.0	3.88–11.2
Mean density (kg m^{-3})	3933–5515	687–1638
Sidereal rotation period (equator)	23.9 h–243 d	9.9 h–17.2 h
Number of known moons	0–2	13–63
Ring systems	no	yes

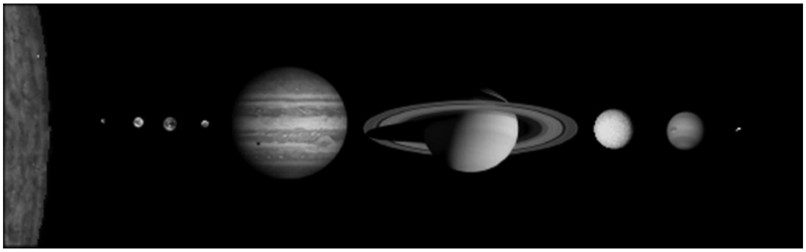

FIGURE 19.1 The relative sizes of the Sun and the planets. From left to right are the Sun, Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, and Pluto (with Charon, one of its moons). A tenth planet, 2003 UB313, which is believed to be slightly larger than Pluto, is not shown in this montage. The distances between objects are not to scale.

(or Earth-like) planets include Mercury, Venus, Earth, and Mars, and the **giant** planets (sometimes called **Jovian**, or Jupiter-like) include Jupiter, Saturn, Uranus, and Neptune. The giant planets are also further separated into the gas giants (Jupiter and Saturn) and the ice giants (Uranus and Neptune). The two major groups have a number of very striking differences, as can be seen by looking through Table 19.1. The relative sizes of the planets and the Sun are illustrated in Fig. 19.1, and specific physical and orbital characteristics of each of the planets can be found in Appendix C.

Many of the differences noted in Table 19.1 are directly related to the distances of the planets from the Sun and their corresponding temperatures. In fact, as we shall see, this temperature effect profoundly influenced the evolution of the terrestrial and giant planets by determining the extent of ice formation in the early solar nebula.

Moons of the Planets

The number of moons orbiting each planet also varies significantly between the terrestrials and the giants. Neither Mercury nor Venus has any moons, Earth has one relatively large moon, and Mars has two tiny satellites. On the other hand, Jupiter, Saturn, Uranus, and

Neptune are known to have at least 63, 47, 27, and 13 moons, respectively. Combined with their ring systems, each of the giant planets possesses a complex orbital system.

With the exception of Pluto[2] and its largest moon, Charon, by far the largest moon in the Solar System *relative to* its parent planet is our own Moon. However, three of the four Galilean moons of Jupiter (Io, Ganymede, and Callisto)[3] and the giant satellite of Saturn (Titan) are physically larger and more massive. In addition, both Ganymede and Titan have radii that are slightly larger than the planet Mercury's even though their masses are somewhat lower.

In some respects, many of the characteristics of the giant moons of the Solar System are similar to those of the terrestrial planets, including active volcanoes on Io and the existence of an atmosphere on Titan. Some of the moons have features unlike anything seen on the planets, however, including the bizarre topography on the surface of Miranda (one of the many moons of Uranus).

The Asteroid Belt

In 1766, before the discoveries of Uranus, Neptune, and Pluto, Johann Titius (1729–1796) uncovered a simple mathematical sequence representing the orbital distances of the planets from the Sun. The sequence was popularized several years later by Johann Elert Bode (1747–1826) and is now known as the **Titius–Bode rule**, or simply *Bode's rule* (see Table 19.2).

When Bode's rule was proposed, it was realized that the rule "predicted" the existence of an object at a distance of 2.8 AU, between the orbits of Mars and Jupiter. It was after a deliberate search that an Italian monk, Giuseppe Piazzi (1746–1826), discovered the first

TABLE 19.2 Predictions of the Titius–Bode Rule. A comparison of the Titius–Bode rule with actual mean orbital distances.

Planet	Titius–Bode Distance (AU)	Actual Mean Distance (AU)
Mercury	$(4 + 3 \times 0\ )/10 = 0.4$	0.39
Venus	$(4 + 3 \times 2^0)/10 = 0.7$	0.72
Earth	$(4 + 3 \times 2^1)/10 = 1.0$	1.00
Mars	$(4 + 3 \times 2^2)/10 = 1.6$	1.52
Ceres	$(4 + 3 \times 2^3)/10 = 2.8$	2.77
Jupiter	$(4 + 3 \times 2^4)/10 = 5.2$	5.20
Saturn	$(4 + 3 \times 2^5)/10 = 10.0$	9.58
Uranus	$(4 + 3 \times 2^6)/10 = 19.6$	19.20
Neptune	$(4 + 3 \times 2^7)/10 = 38.8$	30.05
Pluto	$(4 + 3 \times 2^8)/10 = 77.2$	39.48
2003 UB313	$(4 + 3 \times 2^9)/10 = 154.0$	67

[2] If it were orbiting one of the other planets rather than the Sun, Pluto would be only the eighth largest moon in the Solar System.

[3] Io, Europa, Ganymede, and Callisto were the four moons discovered by Galileo to be orbiting Jupiter; see Section 2.2.

asteroid at approximately that location on January 1, 1801, and named it Ceres, for the patron goddess of Sicily. Today many thousands of asteroids are known, although Ceres is the largest, containing some 30% of the entire mass of the group and having a diameter of roughly 1000 km. Even though there are important exceptions (see Section 22.3), most asteroids orbit the Sun near the ecliptic plane at distances between 2 and 3.5 AU, a region referred to as the **asteroid belt**.

Although Bode's rule agrees reasonably well with the orbits of most of the planets, and it did lead to the "prediction" of Ceres, it fails miserably for objects beyond Uranus. It is widely believed today that Bode's rule is not based on any fundamental physical process and is only a mathematical coincidence. Historically, Bode's rule has often been referred to as Bode's *law*, even though astronomers generally believe today that it is not associated with any basic "law of nature." It is interesting to note, however, that a variation of Bode's rule also works for some of the moons of Jupiter, Saturn, and Uranus. This rephrasing of the rule is explored in Problems 19.2 and 19.3.

While many of the larger moons were certainly formed with their parent planet, others are little more than large rocks that may have been caught in the planet's gravitational field as they wandered by. Many of these rocks are probably captured asteroids.

The Comets and Kuiper Belt Objects

Another important class of objects that orbit our Sun are the **comets**. Once thought to be atmospheric phenomena, and even harbingers of doom,[4] comets are now known to be dirty snowballs of ices and dust. Their spectacular, long tails are simply the escaped dust and gas of the evaporating ball of ice, being driven away from the Sun by radiation pressure and the solar wind. Some comets, like the famous Halley's comet, have relatively short orbital periods of less than 200 years, whereas the long-period comets can take over one million years to orbit the Sun.

From their orbital characteristics, it seems very likely that the present-day source of the short-period comets is the **Kuiper belt**, a collection of icy objects located predominantly near the plane of the ecliptic and beyond the orbit of Neptune, typically ranging from 30 AU to perhaps 1000 AU or more from the Sun. It is now realized that Pluto and its moon Charon, 2003 UB313, Sedna, and Quaoar are among the largest known members of the family of **Kuiper belt objects** (KBOs), also referred to as **Trans-Neptunian Objects** (TNOs). The long-period comets apparently originate in the **Oort cloud**, an approximately spherically symmetric cloud of cometary nuclei with orbital radii of between 3000 and 100,000 AU. Having spent most of their existence in "deep freeze" at the outer reaches of the Solar System, comets and Kuiper belt objects appear to be ancient remnants of its formation, although perhaps not entirely unaffected by nearly 4.6 billion years of exposure to the environment of space.

Meteorites

When asteroids collide with one another, they can produce small fragments known as **meteoroids**. If a meteoroid should happen to enter Earth's atmosphere, the heat generated

[4]Recall the observations of Tycho Brahe, discussed in Section 2.1.

by friction results in a glowing streak across the sky, referred to as a **meteor**. If the rock survives the trip through the atmosphere and strikes the surface, the remnant is known as a **meteorite**. By analyzing the composition of meteorites, we can learn a great deal about the environment in which they originated.

Another source of meteoritic material is the slow disintegration of comets exposed to the heat of the inner Solar System. When Earth encounters the debris left in a comet's orbit, the result is a **meteor shower** of micrometeorites raining down through the planet's atmosphere.

Finally, the dust remaining in orbit about the Sun due to the disintegration of asteroids and comets produces a faint glow from reflected sunlight. Unfortunately, even the lights of a small town are sufficient to obscure this **zodiacal light**.

Solar System Formation: A Brief Overview

All of these features of the Solar System can be understood in terms of its initial formation and subsequent evolution. Our present understanding of Solar System evolution is based on the hypothesis that as the Sun was forming from the gravitational collapse of the original solar nebula, the decreasing radius of the cloud resulted in an increasing rate of spin and the accompanying formation of a disk of material. Within this **accretion disk** the temperature varied with distance from the protosun in such a way that rocks were able to consolidate throughout the disk while ices (primarily water) were able to develop only at distances beyond the outer part of the present-day asteroid belt. As a result, the terrestrial planets accreted from collisions of small preplanetary chunks of material, known as **planetesimals**, that were composed exclusively of rock, while the much larger giants benefited from the additional presence of ices in the planetesimals. The higher temperatures in the inner portions of the disk and the lower masses of the terrestrials also inhibited the capture of lighter gases around those planets, while the cooler, more massive giants were able to accumulate significant and, in the cases of Jupiter and Saturn, very massive primordial atmospheres.

Around the newly formed giant planets, smaller local accretion disks were forming some of the moons seen today. Other moons appeared when planetesimals and fragmented asteroids were captured while wandering through the Solar System. In a different mechanism, it appears that our own Moon was produced when a relatively large planetesimal approximately the size of the present-day Mars collided with the young Earth. After the formation of the planets and their moons, the "rain" of remnant material led to heavy cratering. Although the rate of crater formation has decreased significantly since the time of the early Solar System, the process remains ongoing. Evidence of that violent beginning is still readily apparent on many worlds today.

Most of the icy objects that were drifting among the giant worlds without directly colliding with or being captured by one of the giant planets had their orbits dramatically altered through gravitational interactions. Some of the cometary nuclei passing near Uranus or Neptune were catapulted into much larger orbits, characteristic of the present-day Oort cloud, while those that ventured near Jupiter or Saturn were ejected from the Solar System entirely. Other planetesimals that passed near the giant planets were sent inward to collide with the terrestrial planets or the Sun. The icy bodies that formed beyond the orbit of Neptune remain in that region today, constituting the Kuiper belt. Closer to the Sun, just inside the region of ice formation, rocky remnants of the Solar System's formation still reside in the asteroid belt.

As a direct consequence of tidal and viscous interactions with the accretion disk, along with the scattering of planetesimals, Jupiter migrated closer to the Sun than where it initially began forming, whereas Saturn, Uranus, and particularly Neptune migrated farther from the Sun.

As the discussion of this section indicates, significant progress has been made in understanding the makeup and evolution of our local part of the universe. Throughout the remainder of Part III we will be investigating in more detail the objects that populate our Solar System, together with many of the physical processes that have shaped them. Finally, in Chapter 23 we will reconsider the evolutionary scenario described in the last several paragraphs in light of what we have learned about our Solar System and extrasolar planetary systems.

19.2 ■ TIDAL FORCES

As we saw in Chapter 2, the force of gravity governs the orbits of the planets and their moons in the form of Kepler's laws. In that study we treated those objects as point masses, under the assumption that they are spherically symmetric (see Example 2.2.1). Important consequences arise from relaxing the constraint of spherical symmetry, however. Since one side of a moon is closer to its parent planet than is the opposite side, the planet's gravitational force on a small test mass must be greatest on the moon's near side. This has the effect of elongating the instantaneous shape of the moon. According to Newton's third law, the same situation must also apply to the near and far sides of the planet because of the gravitational influence of its moon. This *differential* force on an object due to its non-zero size is known as a **tidal force**. The resulting nonspherical shapes of the planet and its moon can actually influence their rotation rates by creating torques. If this tidal force is sufficiently great, it is even possible that the smaller world could be disrupted.

The existence of tides on Earth's surface is well known, particularly for those who live near an ocean. There are two high tides approximately every 24 hours, 53 minutes, depending on local coastal features. Less well known are the tidal bulges of the solid Earth, which measure only about 10 cm in height. Since Earth is significantly more massive than the Moon (approximately 81 times), the bulges on the Moon are much larger, resulting in nearly 20 m of deformation at its surface.

The Physics of Tides

To better understand how tides arise on Earth, consider the force on a test mass m_1 located within the planet at a distance r from the Moon's center of mass,

$$F_m = G\frac{Mm_1}{r^2},$$

where M is the mass of the Moon (see Fig. 19.2). Now consider a second mass $m_2 = m_1 = m$, located at a distance dr from m_1 along a line connecting Earth and the Moon. The

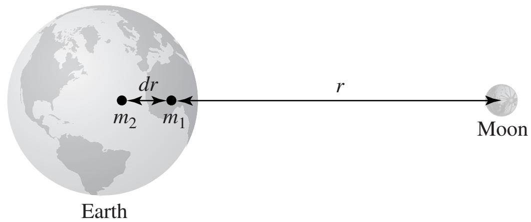

FIGURE 19.2 The tidal force on Earth due to the Moon arises because of the varying values for the Moon's gravitational attraction at different locations inside the planet.

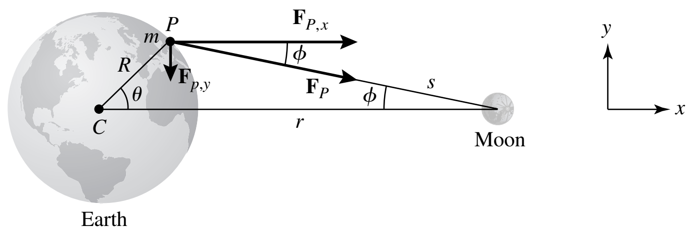

FIGURE 19.3 The geometry of the tidal force acting on Earth due to the Moon.

difference in forces (the differential force) between the two test masses is then

$$\boxed{dF_m = \left(\frac{dF_m}{dr}\right) dr = -2G\frac{Mm}{r^3}\,dr,} \tag{19.1}$$

where dr is taken to be the separation between their centers. Note that the differential force decreases more rapidly with distance than does the force of gravity itself, meaning that the closer the test masses are to the Moon, the more pronounced the effect.

The shape of the tidal bulges on Earth can be understood by analyzing the differences in the gravitational force vectors acting at the center of the planet and at some point on its surface (see Fig. 19.3). For simplicity, we will consider only forces in the x–y plane. Neglecting rotation, the effects are symmetric about the x-axis (the line between the centers of Earth and the Moon). At the center of the planet, the x- and y-components of the gravitational force on a test mass m due to the Moon are given by

$$F_{C,x} = \frac{GMm}{r^2}, \qquad F_{C,y} = 0,$$

while at point P the components are

$$F_{P,x} = \frac{GMm}{s^2}\cos\phi, \qquad F_{P,y} = -\frac{GMm}{s^2}\sin\phi.$$

The differential force between Earth's center and its surface is

$$\Delta\mathbf{F} = \mathbf{F}_P - \mathbf{F}_C = GMm\left(\frac{\cos\phi}{s^2} - \frac{1}{r^2}\right)\hat{\mathbf{i}} - \frac{GMm}{s^2}\sin\phi\,\hat{\mathbf{j}}.$$

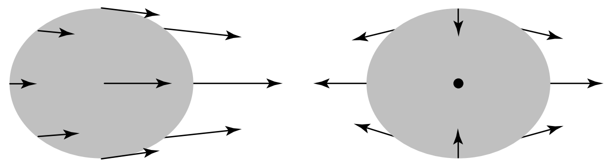

FIGURE 19.4 (a) The gravitational force of the Moon on Earth. (b) The differential gravitational force on Earth, relative to its center.

Next, to simplify the solution somewhat, we write s in terms of r, R, and θ,

$$s^2 = (r - R\cos\theta)^2 + (R\sin\theta)^2 \simeq r^2\left(1 - \frac{2R}{r}\cos\theta\right)$$

where terms of order $R^2/r^2 \ll 1$ have been neglected. Substituting, and recalling that for $x \ll 1$, $(1+x)^{-1} \simeq 1 - x$, we find that the differential force becomes

$$\begin{aligned}\Delta\mathbf{F} \simeq \frac{GMm}{r^2}&\left[\cos\phi\left(1 + \frac{2R}{r}\cos\theta\right) - 1\right]\hat{\mathbf{i}}\\ -\frac{GMm}{r^2}&\left[1 + \frac{2R}{r}\cos\theta\right]\sin\phi\,\hat{\mathbf{j}}.\end{aligned} \tag{19.2}$$

Finally, using the first-order relations $\cos\phi \simeq 1$ and $\sin\phi \simeq (R\sin\theta)/r$, we have

$$\Delta\mathbf{F} \simeq \frac{GMmR}{r^3}\left(2\cos\theta\,\hat{\mathbf{i}} - \sin\theta\,\hat{\mathbf{j}}\right). \tag{19.3}$$

Notice the extra factor of 2 in the x-component when compared with the y-component. You should also compare this result with the expression for the differential force given in Eq. (19.1), noting that here R (the distance between the center and the surface) has replaced dr.

The situation described by Eq. (19.3) is illustrated in Fig. 19.4. The actual gravitational force vectors due to the Moon are directed toward the center of mass of the Moon, but the *differential* force vectors act to compress Earth in the y-direction and elongate it along the line between their centers of mass, producing tidal bulges. It is the symmetry of the bulges that produces two high tides in a 25-hour period as Earth rotates under an orbiting Moon.

The Effects of Tides

In reality, Earth's tidal bulges are not directly aligned with the Moon. This is because the rotation period of Earth is shorter than the Moon's orbital period and frictional forces on the surface of the planet drag the bulge axis ahead of the Earth–Moon line. Because friction is a dissipative force, rotational kinetic energy is constantly being lost and Earth's spin rate is continually decreasing. At the present time, Earth's rotation period is lengthening at the rate of 0.0016 s century^{-1}, which, although slow, is measurable.

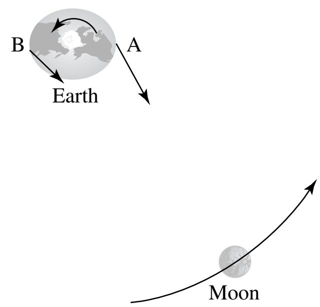

FIGURE 19.5 Earth's bulge A is closer to the Moon than is bulge B, resulting in a net torque on the planet. Note that the diagram is not to scale.

The Moon is also known to be drifting away from Earth by 3 to 4 cm yr^{-1}. The increasing Earth–Moon distance is determined by bouncing laser beams off the mirrors left on the Moon by the Apollo astronauts in the early 1970s and measuring the round-trip light-travel time.

To see how the decrease in Earth's rotation rate and the increase in the Earth–Moon distance are related, we need only consider the torque exerted on Earth by the Moon's interaction with Earth's tidal bulges. In Fig. 19.5, bulge A leads the Moon and is closer to it than is bulge B. As a result, the force exerted on bulge A by the Moon is greater, resulting in a net torque that is slowing Earth's rotation. At the same time, bulge A is pulling the Moon forward, causing the satellite to move farther out. This complementary behavior is just a consequence of the conservation of angular momentum. Neglecting the dynamical influence of the Sun and the other planets on the Earth–Moon system, no external torques exist to alter its total angular momentum. If Earth's rotational angular momentum is decreasing, the orbital angular momentum of the Moon must necessarily increase.

Because of tidal effects, given sufficient time Earth will slow its rotation enough that the same side of the planet will always face the Moon, just as the Moon now keeps the same "face" toward Earth. In the distant future, if inhabitants on Earth's far side want to take romantic moonlight walks, they will need to take vacations halfway around the world. Calculations indicate that this will happen when the length of the day is about 47 current days long.

Synchronous Rotation

In the past the Moon was much closer to Earth than it is today, perhaps taking as little as one week to orbit the planet. It is also probable that the Moon's rotation period was once shorter than its orbital period. The Moon's present **1-to-1 synchronous rotation** is due to the same tidal dissipation that is occurring on Earth today. Its rotational period became synchronized with its orbital period more rapidly than Earth's has, simply because it is much smaller and because Earth produces much larger tidal deformations on the Moon than does the Moon on Earth.

Synchronous rotation is common throughout the Solar System.[5] The two moons of Mars, the four Galilean moons of Jupiter (along with Amalthea, a small moon inside Io's orbit),

[5] Some binary star systems are also known to be in synchronous rotation; see Section 18.1.

and most of the moons of Saturn are in synchronous rotation, as are many of the other moons associated with the outer planets. In addition, Pluto and its largest moon, Charon, have reached the final stage of tidal evolution; they are in mutual synchronous rotation, with the same side of Pluto constantly turned toward the same face of Charon.

An interesting and unusual case of tidal evolution is that of Triton, the giant moon of Neptune. Triton is in synchronous rotation and orbits the planet in a retrograde fashion. In this instance, the tidal bulges in Neptune actually work to cause that moon to spiral *toward* the planet rather than away from it. Apparently it will take billions of years before any catastrophic interaction occurs. On the other hand, Phobos (one of the Martian moons) is in a prograde orbit, but with an orbital period of $7^{h}39^{m}$ that is *shorter* than the rotation period of Mars ($24^{h}37^{m}$). This means that Phobos is inside the planet's **synchronous orbit**, defined to be where the planet's rotation period and the satellite's orbital period are equal.[6] Consequently, it is "outrunning" the tidal bulge axis, and the resulting forces are causing the moon to spiral inward. Phobos's orbit is decaying rapidly enough that if it were to stay intact, it would hit the planet in about 50 million years. The other Martian moon, Deimos, is outside the synchronous orbital radius and is spiraling outward, just as our Moon is.

Additional Tidal Effects from the Sun

Of course, the Earth–Moon system is not in strict isolation. For instance, the Sun also produces tidal forces that act on Earth. When the Sun, Earth, and the Moon are all aligned (at full Moon or new Moon), the differential forces due to the Sun and the Moon add to create unusually large tidal bulges on Earth, called **spring tides**. At first quarter or third quarter, the Sun, Earth, and Moon form a right angle. In this configuration the tides produced by the Sun and the Moon tend to cancel, and unusually low **neap tides** result.

The Roche Limit

It is unlikely that Phobos will actually remain intact long enough to strike the planet. Recall that the differential tidal force is proportional to r^{-3}; as a moon gets closer to its parent planet, tidal effects become more severe. This means that the shape of a moon in synchronous rotation becomes increasingly elongated. Neglecting any internal cohesion forces (i.e., assuming an idealized fluid object), when the orbital distance has decreased sufficiently, it becomes no longer possible to define a shape for the moon such that the force of gravity is perpendicular to the surface at every point. As a result, the surface will continually flow in the direction of the net gravitational force vector. Oscillations will then develop in the extended structure, and the moon will come apart. The maximum orbital radius for which tidal disruption occurs is known as the **Roche limit**, named for Edouard Roche (1820–1883), who first carried out the analysis in 1850. In his study Roche took into consideration orbital and rotational motion, and he assumed a fluid, prolate spheroid (i.e., a football-shaped moon).

To make an order-of-magnitude estimate of the orbital radius at which a moon will break apart, assume (incorrectly) that this happens when the differential force exceeds the

[6]Earth's synchronous orbit is sometimes referred to as geosynchronous orbit. Artificial satellites placed in geosynchronous equatorial orbits remain fixed over the same geographic point on the surface. Communications satellites are generally placed in such orbits.

self-gravitational force holding the moon together. Furthermore, assume for simplicity that the moon and the planet are spherical, and neglect any centrifugal effects. In this case, if the moon is to be tidally disrupted, the inward gravitational acceleration produced by the moon at a point located on its surface closest to the planet must be smaller than the outward differential gravitational acceleration produced by the planet, or

$$\frac{GM_m}{R_m^2} < \frac{2GM_pR_m}{r^3},$$

where M_p and M_m are the masses of the planet and moon, respectively, R_m is the radius of the moon, and r is the distance between the centers of the two worlds. Substituting $M_p = 4\pi R_p^3\overline{\rho}_p/3$ and $M_m = 4\pi R_m^3\overline{\rho}_m/3$, where $\overline{\rho}_p$ and $\overline{\rho}_m$ are the average densities of the planet and moon, respectively, and solving for r, we find that a moon will be tidally disrupted if its orbit is less than

$$\boxed{r < f_R\left(\frac{\overline{\rho}_p}{\overline{\rho}_m}\right)^{1/3} R_p,} \tag{19.4}$$

where, in our case, $f_R = 2^{1/3} = 1.3$. In his more careful analysis, Roche found a larger value for the leading constant of $f_R = 2.456$. The fact that our result gave too small a value for the radius reflects the incorrect assumption that it is the differential force exceeding self-gravity that is ultimately responsible for the disintegration of the satellite. Since oscillations in the body will develop at greater radii, self-gravity is still significantly greater than the differential term at the true Roche limit. (Although not considered in this analysis, self-cohesion of an object that is provided by the electromagnetic force, such as molecular bonds or the formation of a crystal lattice, can also decrease the point at which an object will be disrupted.)

Example 19.2.1. The average density of Saturn is 687 kg m^{-3} and its planetary radius is 6.03×10^7 m. Using a value of $f_R = 2.456$, the Roche limit for a moon having an average density of 1200 kg m^{-3} is 1.23×10^8 m. Much of the ring system of Saturn lies within this orbital radius given by the Roche limit, and all of Saturn's large moons are farther out. The material within ring systems may be the result of disintegrating or tidally disrupted moons that wandered within the Roche limit.

19.3 ■ THE PHYSICS OF ATMOSPHERES

Our Solar System today is the result of billions of years of ongoing evolution caused by a host of physical processes. Subtle differences in initial conditions of neighboring planets have led to the very different worlds we see today. We will discuss some of the more frequently encountered atmospheric processes in this section and then, in later chapters, describe the unique characteristics of each planet.

The Temperatures of the Planets

As has already been mentioned, the temperatures of the planets played a key role in their formation and evolution. During the formation stage, the temperature structure of the solar nebula influenced whether a planet would become a terrestrial or a giant; this point will be discussed more fully in Section 23.2. Temperature also helped to determine the current composition of each planet's atmosphere.

The Stefan–Boltzmann equation (Eq. 3.17) is the most significant factor in determining the present-day temperatures of the planets in the Solar System. Under equilibrium conditions, a planet's total energy content must remain constant. Therefore, all of the energy absorbed by the planet must be re-emitted; if this were not so, the planet's temperature would change with time.

To estimate a planet's equilibrium temperature, assume that the planet is a spherical blackbody of radius R_p and temperature T_p in a circular orbit a distance D away from the Sun. For simplicity, we will assume that the planet's temperature is uniform over its surface[7] and that the planet reflects a fraction a of the incoming sunlight (a is known as the planet's **albedo**). From the condition of thermal equilibrium, the sunlight that is not reflected must be absorbed by the planet and subsequently re-emitted as blackbody radiation. Of course, we will also treat the Sun as a spherical blackbody having an effective temperature $T_\odot = T_e$ and radius $R_\odot$. It is left as an exercise to show that the temperature of the planet is given by

$$\boxed{T_p = T_\odot(1-a)^{1/4}\sqrt{\frac{R_\odot}{2D}}.} \tag{19.5}$$

Note that the temperature of the planet is proportional to the effective temperature of the Sun and does not depend on the size of the planet.

Example 19.3.1. Using Earth's average value of $a = 0.3$ in Eq. (19.5), the temperature of a blackbody Earth is

$$T_\oplus = 255\ \text{K} = -19^\circ\text{C} = -1^\circ\text{F}.$$

This value is substantially below the freezing point of water and (fortunately!) is not the correct temperature at the surface of the planet. This analysis neglected the **greenhouse effect**, a significant warming due largely to the water vapor in Earth's atmosphere.[8] According to Wien's law (Eq. 3.15), Earth's blackbody radiation is emitted primarily at infrared wavelengths. This infrared radiation is absorbed and then re-emitted by the atmospheric greenhouse gases, which act as a thermal blanket to warm Earth's surface by about 34°C. A simple model of the greenhouse effect is explored in Problem 19.13. Greenhouse warming on Venus has been much more dramatic; the reasons for this will be discussed in Section 20.2.

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[7]This assumption is a reasonable approximation if the planet is rapidly rotating or has a circulating atmosphere.
[8]Carbon dioxide, methane, and chlorofluorocarbons also contribute to the greenhouse effect.

The Chemical Evolution of Planetary Atmospheres

The evolution of a planetary atmosphere is a complex process that depends on the local temperature of the solar nebula during the time of the planet's formation, together with the planet's temperature, gravity, and local chemistry following the formation process. In the case of the terrestrial planets, outgassing from rocks and volcanos also played a role after the development of the initial, primordial atmosphere. On Earth, the development of life has also contributed significantly to the evolution of its atmosphere. Impacting comets and meteorites affect planetary atmospheres as well.

A critical component in the development of an atmosphere is the ability of the planet to retain specific atoms or molecules. Recall from the discussion in Section 8.1 that for a gas in thermal equilibrium, the number of particles having velocities between v and $v + dv$ is given by the Maxwell–Boltzmann velocity distribution (see Eq. 8.1 and Fig. 8.6). At some critical height in an atmosphere, when the number density is low enough that collisions among gas particles become negligible, particles moving upward will travel only under the influence of gravity, following trajectories described by simple projectile motion. Those atoms or molecules that are not moving rapidly enough to escape, or that do not have the correct trajectories, will fall back down into the denser layers and undergo collisions with the gas. On the other hand, particles that are moving upward and have velocities that are sufficiently great will be able to escape the gravitational pull of the planet altogether and move out into interplanetary space. It is this process that can allow the atmospheres of some planets (or at least specific chemical components of those atmospheres) to "leak off." The region in an atmosphere where the mean free path of the particles becomes long enough for them to travel without appreciable collisions is referred to as the **exosphere**.

Because of the high-velocity tail of the Maxwell–Boltzmann distribution, and because of the amount of time that has elapsed since the Solar System formed, if a particular component of the atmosphere is going to escape, it is not necessary that the root-mean-square average velocity of those particles be greater than the escape speed. It is only necessary that a sufficiently large number of particles have speeds greater than $v_{\rm esc}$. As a rough estimate, a planet will have lost a particular component of its atmosphere by now if, for that component (either molecular or atomic),

$$v_{\rm rms} > \frac{1}{6} v_{\rm esc}.$$

Using Eqs. (2.17) and (8.3) for the escape and root-mean-square velocities, respectively, the temperature required for a gas of particles of mass m to escape a planet of mass M_p and radius R_p is approximately

$$T_{\rm esc} > \frac{1}{54} \frac{G M_p m}{k R_p}. \tag{19.6}$$

Example 19.3.2. Earth has an atmosphere composed of approximately 78% N_2 and 21% O_2 by number, while the Moon, which is on average the same distance from the Sun, has no significant atmosphere. From Example 19.3.1, the blackbody equilibrium temperature of an *airless* Earth should be 255 K. Since the Moon's albedo is only 0.07, its blackbody temperature is somewhat higher (274 K). In reality, the vertical temperature structure of Earth's atmosphere is very complex and depends on its hydrodynamic motions together with the ability of various atoms and molecules to absorb radiation. Near the top of the atmosphere, the temperature is also strongly dependent on the amount of solar activity. Within the exosphere the characteristic temperature is about 1000 K.

Consider Earth's ability and the Moon's inability to retain molecular nitrogen. The mass of an N_2 molecule is approximately 28 u $= 4.7 \times 10^{-26}$ kg, the mass and radius of Earth are 5.9736×10^{24} kg and 6.378136×10^6 m, respectively, and the mass and radius of the Moon are 7.349×10^{22} kg and 1.7371×10^6 m, respectively (see Appendix C). The temperatures required for the nitrogen to escape from each world can now be estimated from Eq. (19.6), giving $T_{\text{esc},\oplus} > 3900$ K and $T_{\text{esc,Moon}} > 180$ K. Since Earth's exospheric temperature is cooler and the Moon is warmer than these values, Earth has been able to retain its molecular nitrogen, whereas the Moon could not.

Since O_2 is more massive (32 u), even higher temperatures are required for that molecule to escape.

The Loss of Atmospheric Constituents

The loss of specific components of an atmosphere can be understood in more detail by appealing directly to Eq. 8.1, the Maxwell–Boltzmann distribution, $n_v\, dv$. As particles move about randomly in the gas, some of them are traveling approximately vertically upward and therefore have the best chance of escaping. The number of particles with velocities between v and $v + dv$ passing through a horizontal slab of cross-sectional area A and vertical thickness dz during a time interval dt is given by

$$dN_v\, dv = (n_v\, dV)\, dv = A\, dz\, n_v\, dv = Av_z\, dt\, n_v\, dv = C_g Av\, dt\, n_v\, dv,$$

where C_g is a geometrical factor that takes into consideration the requirement that, of all the velocity components of the randomly moving particles, only positive vertical components will be considered. Dividing through by the time interval, we obtain the *rate* at which particles with velocities between v and $v + dv$ are crossing the surface. Furthermore, if we assume that the atmosphere is spherical at the location of the exosphere, so that $A = 4\pi R^2$, then the number of particles per second with speeds between v and $v + dv$ moving vertically upward through the entire exosphere is given by

$$\dot{N}_v\, dv \equiv \frac{dN_v}{dt}\, dv = 4\pi R^2 C_g v n_v\, dv.$$

Finally, to determine the number of particles per second leaving the atmosphere, it is necessary only to consider those particles with sufficiently high velocities, namely $v > v_{\text{esc}}$. Substituting Eq. (8.1) and integrating, we have

$$\dot{N} = \frac{n\pi R^2}{4}\left(\frac{m}{2\pi kT}\right)^{3/2}\int_{v_{\text{esc}}}^{\infty} 4\pi v^3 e^{-mv^2/2kT}\, dv, \tag{19.7}$$

where C_g has been set equal to $1/16$, based on a careful analysis of the geometry of the problem. In Problem 19.16 you will be asked to show that at some height z in the atmosphere, where the particle number density is $n(z)$, Eq. (19.7) reduces to

$$\dot{N}(z) = 4\pi R^2 \nu n(z), \tag{19.8}$$

where

$$\nu \equiv \frac{1}{8}\left(\frac{m}{2\pi kT}\right)^{1/2}\left(v_{\rm esc}^2 + \frac{2kT}{m}\right)e^{-mv_{\rm esc}^2/2kT} \tag{19.9}$$

is an **atmospheric escape parameter** that has units of velocity.

ν describes the rate at which gas particles of mass m escape across a unit area for a specified number density $n(z)$ in the exosphere. The atmospheric escape parameter can also be thought of as the *effective thickness* of the atmosphere of a certain species that evaporates away (or "leaks off") per second. In Fig. 19.6, $\log_{10}\nu$ is plotted as a function of the mass of specific components in Earth's atmosphere, where a temperature of 1000 K has been used, characteristic of a mean value in the exosphere. For comparison, $\log_{10}\nu$ has also been plotted for the same species using the Moon's escape velocity and a typical temperature of 274 K. Note that of the components listed, only molecular hydrogen and helium have essentially completely escaped Earth's atmosphere, whereas the Moon has lost all of its atmosphere, including the heavier molecules listed.

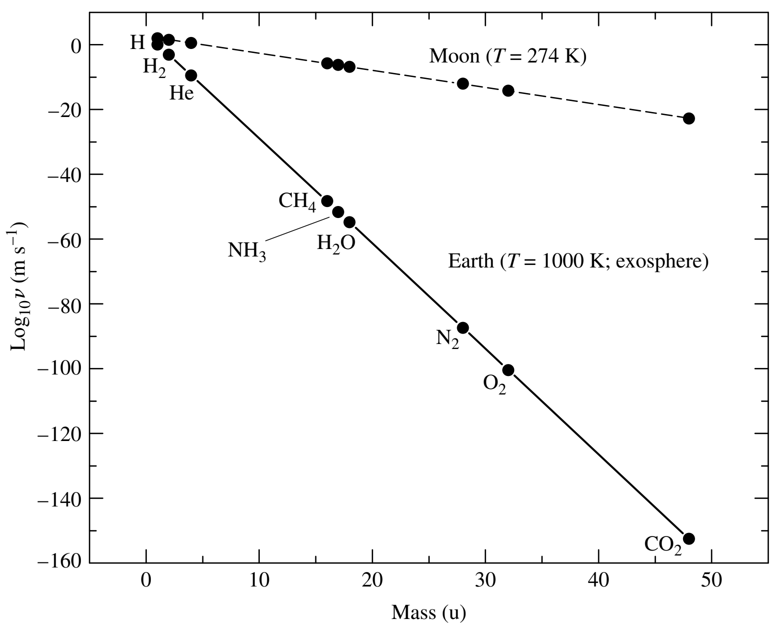

FIGURE 19.6 The logarithm of the atmospheric escape parameter, ν, as a function of atomic weight for various chemical species in Earth's atmosphere and on the surface of the Moon. Note that Earth has lost most of its atomic and molecular hydrogen and its helium, while retaining the other molecules listed. The Moon has lost all of its atmosphere.

Example 19.3.3. Equations (19.8) and (19.9) can be used to estimate the amount of time required for molecular nitrogen to escape from Earth's atmosphere. A rough calculation of the total number of N_2 molecules in the atmosphere can be made by assuming that the number density decreases approximately exponentially with height, just as the pressure does if the atmosphere is nearly isothermal (see Section 10.4).[9] Then

$$n(z) = n_0 e^{-z/H_P}$$

where n_0 is the number density at the surface and H_P is the pressure scale height, given by Eq. (10.70). Using the ideal gas law (Eq. 10.11) with the mass of the nitrogen molecule (M_{N_2}) being used for μm_H, the pressure scale height may be written as

$$H_P = \frac{P}{\rho g} = \frac{kT}{g M_{N_2}}. \tag{19.10}$$

Notice that the pressure scale height is different for particles of different masses. Using characteristic values at Earth's surface ($T = 288$ K and $g = 9.80$ m s^{-2}), $H_P = 8.7$ km for molecular nitrogen.

Making the rough assumption that the pressure scale height remains constant with altitude, and neglecting the slight change in r relative to Earth's surface, the number density can now be integrated over the volume of the atmosphere, giving

$$N = 4\pi R_\oplus^2 n_0 H_P.$$

Taking the number density of nitrogen molecules near the surface to be $n_0 = 2 \times 10^{25}$ m^{-3}, the total number of nitrogen molecules in the atmosphere is $N = 9 \times 10^{43}$.

From Eq. (19.9) and Fig. 19.6, the atmospheric escape parameter for nitrogen molecules from Earth's exosphere is $\nu = 4 \times 10^{-88}$ m s^{-1}. Also, at the height of the exosphere (approximately 500 km), the mean number density of N_2 is 2×10^{11} m^{-3}. Using Eq. (19.8), we find that the rate at which nitrogen molecules are escaping Earth's atmosphere is approximately $\dot{N} = 4 \times 10^{-62}$ s^{-1}. Dividing the total number of available molecules by the rate of loss, the time required to dissipate the nitrogen in Earth's atmosphere is estimated to be

$$t_{N_2} = \frac{N}{\dot{N}} = 2 \times 10^{105} \text{ s} = 6 \times 10^{97} \text{ yr}.$$

It is safe to say that Earth's atmospheric nitrogen is not going to escape any time soon!

The situation is very different for atomic hydrogen in Earth's atmosphere, however. This case will be considered in detail in Problems 19.14 and 19.19.

[9]Earth's atmosphere actually differs appreciably from an isothermal approximation. As a result, the estimate of $n(z)$ used here would need to be modified significantly in a more careful analysis. Nevertheless, this "back-of-the-envelope" calculation illustrates many of the basic physical principles involved and yields the correct general conclusion.

Besides the loss of high-velocity particles from the exponential tail of the Maxwell–Boltzmann distribution, other factors also contribute to the dissipation of an atmosphere. Molecular **photodissociation**, caused by the absorption of UV photons in the upper atmosphere, breaks down some molecules into atoms or lighter molecules, with the result that the individual particles have greater speeds (recall the expression for the root-mean-square speed, Eq. 8.3). For instance, $H_2 + \gamma \rightarrow H + H$. The solar wind can also contribute to the loss of particles through collisions in the upper atmosphere, causing direct ejection, or molecular dissociation and subsequent escape. Even heating caused by impacting meteorites and comets can accelerate the loss of atmospheric constituents.

Gravitational Separation of Atmospheric Constituents

Another feature of atmospheric physics that can affect the loss of certain components is the **gravitational separation** (also known as **chemical differentiation**) of constituents of an atmosphere by weight. In the absence of the continual mixing caused by convection at lower altitudes, composition differences develop with height in the upper atmosphere. This effect can be understood by referring to the expression for the pressure scale height in the form of Eq. (19.10). For a given temperature, the pressure scale height increases as the mass decreases, meaning that the number densities of lighter particles do not diminish as rapidly with z. As a result, lighter particles become relatively more abundant in the upper atmosphere, enhancing the likelihood of their escape.

Circulation Patterns

As is the case with stars (see Section 10.4), convection in planetary atmospheres is driven largely by steep temperature gradients. Near the equator, where the intensity of the sunlight is greatest, the atmosphere heats up and the warm gas rises. The gas then migrates to cooler regions at high latitudes, where it sinks back down again. The cycle closes when the gas returns to the warmer regions near the equator. If the warm air were able to migrate all the way from the equator to the poles before sinking, the global pattern illustrated in Fig. 19.7(a) would occur. This hypothetical circulation pattern is known as **Hadley circulation**.

In reality, the warmer air at higher altitudes is undergoing radiative cooling as it migrates toward the poles. At about 30° N and S latitude, the air has given up enough heat that it sinks and returns to the equator, where it is reheated again. Similarly, the colder air that is migrating from the poles toward the equator at lower altitudes heats and rises at about 55° N and S latitude, returning to the poles where it sinks again. This breaks up the global Hadley circulation pattern into three zonal components, as shown in Fig. 19.7(b).

These general zonal weather patterns are further complicated by the planet's rotation. Since a rotating body does not constitute an inertial reference frame, pseudo-forces, such as the **Coriolis force**, are present.

Assume for simplicity that Earth is perfectly spherical. At a latitude L, a point on the surface is located a distance r_L from the rotation axis, given by

$$r_L = R_\oplus \cos L.$$

Letting the angular rotation speed of Earth be ω, the eastward speed of the surface at the